
Working title: Parameterizations in target trial emulations

Exact order to be determined.

October 2, 2026

ABSTRACT

1 INTRODUCTION

This note is about parameterizations in target trial emulation models. Specifically, we will discuss about ways to parameterize counterfactual contrasts under different regimes in sequential trials. The note is two-fold. First, we clarify which causal contrasts are coherent across sequential trials. That is, we find regimes we can compare without concerns on incoherent modeling. We show that not all regimes are coherent and find the set of coherent causal contrasts. Within this coherent subset, we additionally find the set that can be parameterized using structural nested mean models (SNMM). We note that doubly-robust g-estimators are only applicable to SNMM but not any coherent causal contrasts, we propose alternative estimators that can be used for estimation and be more efficient than the traditional inverse-probability weighting (IPW) estimators.

This question is motivated by the overparameterization problem in history-adjusted marginal structural models (HA-MSMs). Consider data collected at two treatment occasions, (L_0, A_0, L_1, A_1, Y) , where L_0, A_0, L_1, A_1 are binary and Y is normally distributed conditional on these variables. We parameterize the distribution P using the following saturated structural causal model (SCM):

$$\begin{aligned}
L_0 &\sim \text{Bern}(\text{expit}\{\alpha\}), \\
A_0 \mid L_0 &\sim \text{Bern}(\text{expit}\{\beta_0 + \beta_1 L_0\}), \\
L_1 \mid L_0, A_0 &\sim \text{Bern}(\text{expit}\{\gamma_0 + \gamma_1 L_0 + \gamma_2 A_0 + \gamma_3 L_0 A_0\}), \\
A_1 \mid L_0, A_0, L_1 &\sim \text{Bern}(\text{expit}\{\zeta_0 + \zeta_1 L_0 + \zeta_2 A_0 + \zeta_3 L_1 + \zeta_4 L_0 A_0 \\
&\quad + \zeta_5 L_0 L_1 + \zeta_6 A_0 L_1 + \zeta_7 L_0 A_0 L_1\}), \\
Y &= \xi_0 + \xi_1 L_0 + \xi_2 A_0 + \xi_3 L_1 + \xi_4 A_1 \\
&\quad + \xi_5 L_0 A_0 + \xi_6 L_0 L_1 + \xi_7 L_0 A_1 + \xi_8 A_0 L_1 + \xi_9 A_0 A_1 + \xi_{10} L_1 A_1 \\
&\quad + \xi_{11} L_0 A_0 L_1 + \xi_{12} L_0 A_0 A_1 + \xi_{13} L_0 L_1 A_1 + \xi_{14} A_0 L_1 A_1 \\
&\quad + \xi_{15} L_0 A_0 L_1 A_1 + \varepsilon,
\end{aligned}$$

where $\varepsilon \sim N(0, \sigma^2)$ is independent of (L_0, A_0, L_1, A_1) . The distribution P is therefore parame-

terized as $P \equiv P(\alpha, \beta, \gamma, \zeta, \xi, \sigma^2)$, with $\alpha \in \mathbb{R}$, $\beta \in \mathbb{R}^2$, $\gamma \in \mathbb{R}^4$, $\zeta \in \mathbb{R}^8$, $\xi \in \mathbb{R}^{16}$, and $\sigma^2 > 0$. Throughout, we assume consistency, sequential positivity, and sequential exchangeability. Under these assumptions, the counterfactual means considered below are identified from the observed-data distribution P through the g -formula. For this two-time setting, a saturated HA-MSM can be specified as follows:

$$\begin{aligned} \mathbb{E}_P(Y^{a_0, a_1} \mid L_0 = l_0) &= \theta_0^0 + \theta_1^0 l_0 + \theta_2^0 a_0 + \theta_3^0 a_1 \\ &+ \theta_4^0 l_0 a_0 + \theta_5^0 l_0 a_1 + \theta_6^0 a_0 a_1 + \theta_7^0 l_0 a_0 a_1, \\ \mathbb{E}_P(Y^{A_0, a_1} \mid L_0 = l_0, A_0 = a_0, L_1 = l_1) &= \theta_0^1 + \theta_1^1 l_0 + \theta_2^1 a_0 + \theta_3^1 l_1 + \theta_4^1 a_1 \\ &+ \theta_5^1 l_0 a_0 + \theta_6^1 l_0 l_1 + \theta_7^1 l_0 a_1 + \theta_8^1 a_0 l_1 + \theta_9^1 a_0 a_1 + \theta_{10}^1 l_1 a_1 \\ &+ \theta_{11}^1 l_0 a_0 l_1 + \theta_{12}^1 l_0 a_0 a_1 + \theta_{13}^1 l_0 l_1 a_1 + \theta_{14}^1 a_0 l_1 a_1 + \theta_{15}^1 l_0 a_0 l_1 a_1. \end{aligned}$$

Saturated models ensure that the correct specifications of the conditional counterfactual means. However, their parameters may still be subject to cross-trial compatibility constraints and lead to variation-dependent causal contrasts. To illustrate this issue, consider the following three causal contrasts,

$$\begin{aligned} \mathbb{E}_P(Y^{0,1} - Y^{0,0} \mid L_0 = 0) &= \theta_3^0, \\ \mathbb{E}_P(Y^{0,1} - Y^{0,0} \mid L_0 = 0, A_0 = 0, L_1 = 0) &= \theta_4^1, \\ \mathbb{E}_P(Y^{0,1} - Y^{0,0} \mid L_0 = 0, A_0 = 0, L_1 = 1) &= \theta_4^1 + \theta_{10}^1, \end{aligned}$$

where θ_3^0 is a causal contrast in Trial 0 among individuals with $L_0 = 0$. The parameters θ_4^1 and $\theta_4^1 + \theta_{10}^1$ are causal contrasts in Trial 1 among individuals with histories $(L_0, A_0, L_1) = (0, 0, 0)$ and $(0, 0, 1)$, respectively. To specify these contrasts freely across trials, we would require θ_3^0 , θ_4^1 , and θ_{10}^1 to be jointly variation-independent: each parameter could take any value in its marginal parameter space without restricting the others. However, the HA-MSM parameters in this example

do not satisfy this property. Applying the g -formula under P yields

$$\theta_4^1 = \xi_4, \quad \theta_{10}^1 = \xi_{10}, \quad \theta_3^0 = \xi_4 + \xi_{10} \expit(\gamma_0).$$

Consequently, there is a constrain on the HA-MSM parameters

$$\theta_3^0 = \theta_4^1 + \theta_{10}^1 \expit(\gamma_0) \tag{1}$$

Figure 1 illustrates this constraint and show that $(\theta_3^0, \theta_4^1, \theta_{10}^1)$ can only lie in the hyperplane. Because $0 < \expit(\gamma_0) < 1$, the combination $\theta_4^1 = 1$, $\theta_{10}^1 = -1$, and $\theta_3^0 = 1$, represented by the red vertex in the figure, is impossible. Thus, these three parameters cannot vary freely.

When the trial-specific HA-MSMs are fitted separately using IPW estimators, their estimates need not satisfy the above constraint in finite samples. For example, suppose that $\hat{\theta}_4^1 > 0$ and $\hat{\theta}_{10}^1 > 0$, with both coefficients significant at the 5% level. The estimated effect of A_1 is then positive in both L_1 strata among individuals with $L_0 = A_0 = 0$. Under sequential exchangeability,

$$\theta_3^0 = \mathbb{E}_P(Y^{0,1} - Y^{0,0} \mid L_0 = 0, A_0 = 0),$$

so θ_3^0 also represents the effect of A_1 among individuals with $L_0 = A_0 = 0$, averaged over L_1 . A negative estimate $\hat{\theta}_3^0$ would therefore be incompatible with the two positive stratum-specific estimates, yielding contradictory estimated directions of effect in that population. Additionally, a more common disagreement arises when the Trial 0 analysis fails to reject the null hypothesis $\theta_3^0 = 0$, despite evidence of positive effects in Trial 1. Although a failure to reject the null does not establish a null effect, this difference in statistical significance can complicate the interpretation of findings across trials. Table 1 reports the frequencies of these disagreements in simulations. Even with 16000 samples with all models to be correct, researchers may still get inconsistent conclusion in 28.5% of the time. Furthermore, with 2000 to 4000 samples, researchers still have a chance of 0.1% to get contradictory results. These logic error arise not because of any incorrect models

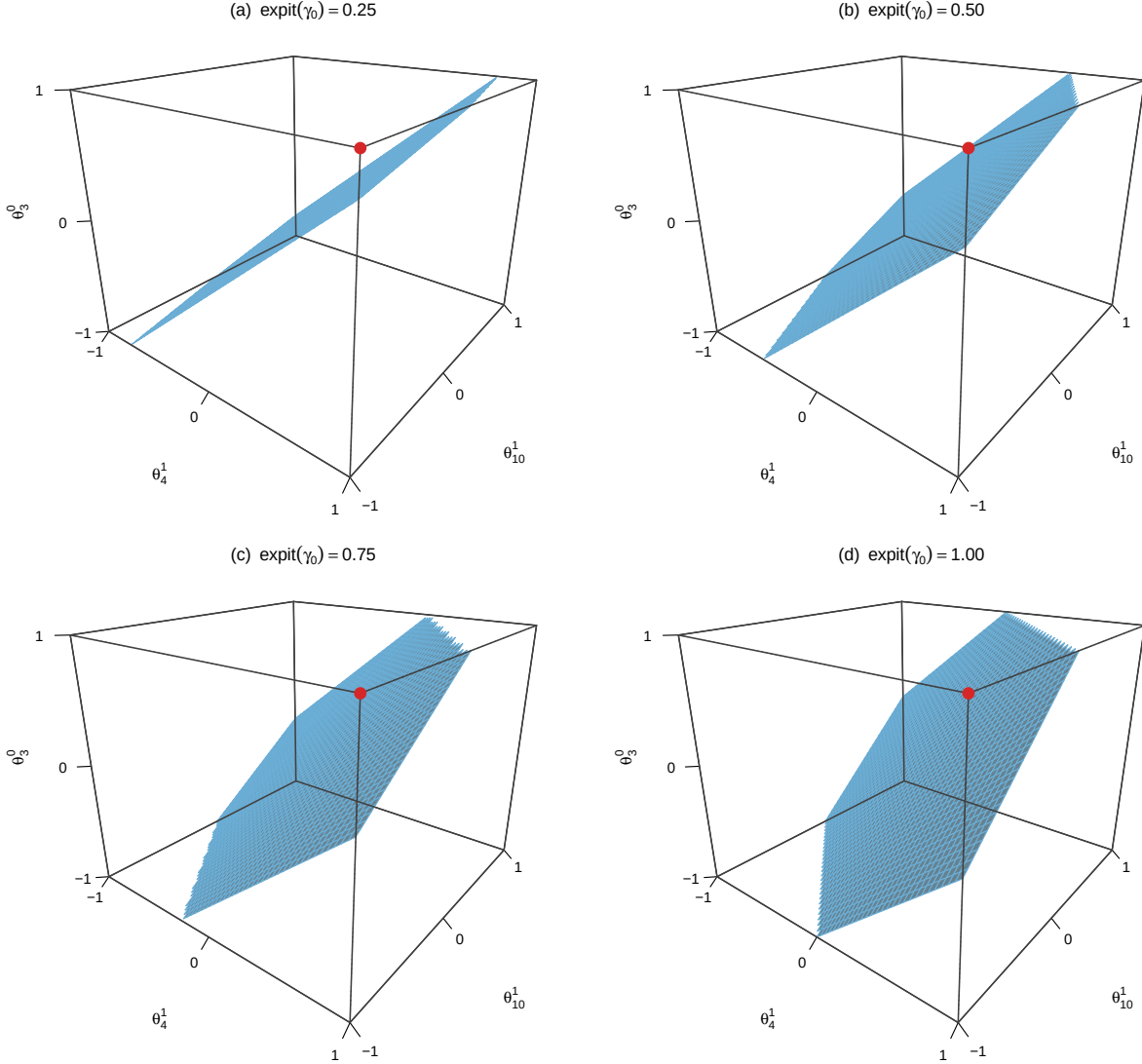


Figure 1: Caption

but because of overparamaterization. Given the valuable data sources, statisticians may pursue to use data to estimate parameters efficiently, and overparameterization may be a waste of data and is preferred to be avoided.

In sequential trials, [Guo et al. \(2021\)](#); [Robins \(2004\)](#); [Robins et al. \(2000\)](#) have shown that SNMMs provide a coherent parameterization of causal contrasts. In the setting considered here, causal contrasts that admit an SNMM representation are variation-independent. This raises the question of whether an SNMM representation is also necessary for variation-independence, such that contrasts without this representation must be variation-dependent. We show that variation in-

Table 1: Inconsistent conclusions in the two-time example. “Both” means $\hat{\theta}_4^1 > 0$ and $\hat{\theta}_{10}^1 > 0$, each significant at the 5% level, so Trial 1 reports that A_1 is beneficial whatever the value of L_1 . The identity $\theta_3^0 = \theta_4^1 + \theta_{10}^1 \text{expit}(\gamma_0)$ then forces $\theta_3^0 > 0$, so a nonsignificant or significantly negative $\hat{\theta}_3^0$ contradicts Trial 1 on the same population. Truth: $\theta_4^1 = 0.25$, $\theta_{10}^1 = 0.50$, $\text{expit}(\gamma_0) = 0.50$, $\theta_3^0 = 0.50$; 1000 replicates per sample size, bootstrap $B = 300$.

n	Trial 0 contradicts Trial 1		
	Both significant	Both significant, $\hat{\theta}_3^0$ non-significant	Both significant, $\hat{\theta}_3^0 < 0$ significant
1,000	3.7%	3.2%	0.00%
2,000	20.4%	16.3%	0.10%
4,000	68.6%	52.5%	0.10%
8,000	96.4%	59.9%	0.00%
16,000	100.0%	28.5%	0.00%

dependence requires a weaker condition than an SNMM representation. Specifically, we establish two main results: (1) a sufficient condition for a collection of causal contrasts to be variation-independent, and (2) a sufficient condition for such a collection to admit an SNMM representation.

2 NOTATION AND MODELS

Let O denote all variables collected during the study: $O \equiv (Y_0, L_0, A_0, L_1, A_1, \dots, L_K, A_K, Y_{K+1})$, and define the full data as $X \equiv (O, \{Y_{K+1}^{\bar{a}_K} : \bar{a}_K \in \{0, 1\}^{K+1}\})$, where $Y_{K+1}^{\bar{a}_K}$ denotes the counterfactual outcome for Y_{K+1} under treatment history $\bar{a}_K$. Thus, X consists of the observed data together with all counterfactual outcomes under static treatment sequences. We let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P$, where $P \in \mathcal{P}$ is a full-data law satisfying the regularity conditions. For any integrable function f , we use $\mathbb{E}_P(f)$ to denote $\int_x f(x) dP(x)$. By convention, we use overbars to denote histories; for example, $\bar{Z}_k = (Z_0, Z_1, \dots, Z_k)$. Similarly, we use underbars to denote future sequences; for example, $\underline{Z}_k = (Z_k, Z_{k+1}, \dots, Z_K)$. We use $\bar{S}_m = (\bar{L}_m, \bar{A}_{m-1})$ to denote the history before treatment A_m , for $m = 0, \dots, K$, with the conventions $\bar{A}_{-1} = \emptyset$ and $\bar{a}_{-1} = \emptyset$. We use $\bar{S}_j(P)$ to denote the support of $\bar{S}_j$ under P , and assume for each $m = 0, \dots, K$, $\bar{S}_m$ has a common support over $P \in \mathcal{P}$, denoted by $\bar{S}_m$. We define a treatment arm to be a collection of functions that assign 0 or 1 based on the covariate history. For a two-arm trial starting from time t_m , the two treatment

arms are defined as

$$\{\bar{A}_{m-1}, g_m^m(\bar{S}_m), g_{m+1}^m(\bar{S}_{m+1}), \dots, g_K^m(\bar{S}_K)\}$$

and

$$\{\bar{A}_{m-1}, g_m^{m,'}(\bar{S}_m), g_{m+1}^{m,'}(\bar{S}_{m+1}), \dots, g_K^{m,'}(\bar{S}_K)\},$$

where $g_k^m(\bar{s}_k), g_k^{m,'}(\bar{s}_k) \in \{0, 1\}$ denote the treatment values participants should follow at time t_k in Trial m under the two treatment arms, for $0 \leq m \leq k \leq K$. For notational simplicity, we call the two arms the g^m -arm and the $g^{m,'}$ -arm and write them as

$$\underline{g}_m^m = \{g_m^m, g_{m+1}^m, \dots, g_K^m\}, \quad \underline{g}_m^{m,'} = \{g_m^{m,'}, g_{m+1}^{m,'}, \dots, g_K^{m,'}\}.$$

We call the trial starting from time t_m Trial m . The counterfactual outcome under the g^m -arm can be understood as a measurable function of the full data X by the following map,

$$Y_{K+1}^{\bar{A}_{m-1}, \underline{g}_m^m} \equiv \sum_{\bar{a}_K \in \{0,1\}^{K+1}} \mathbb{I}(\bar{A}_{m-1} = \bar{a}_{m-1}) \prod_{j=m}^K \mathbb{I}\{a_j = g_j^m(\bar{S}_j)\} Y_{K+1}^{\bar{a}_K}.$$

Thus, the law of $Y_{K+1}^{\bar{A}_{m-1}, \underline{g}_m^m}$ is the push-forward of the full-data law P under this function. Since the treatment functions may be chosen arbitrarily within the class considered above, the same construction applies to every treatment arm with pre-specified arm functions; e.g., the counterfactual outcome under the $g^{m,'}$ -arm can be defined by replacing g_j^m with $g_j^{m,'}$ in the above definition. Consequently, the full-data law P determines the law of the counterfactual outcome under every such treatment arm.

Under this framework, we allow for different treatment arms under different trials. For example, in the case of $K = 2$, there are three trials beginning at t_0 , t_1 , and t_2 . The counterfactual outcomes of contrast are (1) $Y_3^{g_0^0, g_1^0, g_2^0}$ and $Y_3^{g_0^{0,'}, g_1^{0,'}, g_2^{0,'}}$; (2) $Y_3^{a_0, g_1^1, g_2^1}$ and $Y_3^{a_0, g_1^{1,'}, g_2^{1,'}}$, for $a_0 = 0, 1$; and (3) $Y_3^{a_0, a_1, g_2^2}$ and $Y_3^{a_0, a_1, g_2^{2,'}}$, for $a_0 = 0, 1$, $a_1 = 0, 1$. For real-valued outcomes, we consider conditional mean contrasts between the two treatment arms of each trial, which are usually

treated as conditional average treatment effects, and thus, functions for time-varying covariates. For example, when $K = 2$, given a full-data law P for X , the conditional causal contrasts can be expressed,

1. $\mathbb{E}_P \left(Y_3^{g_0^0, g_1^0, g_2^0} \mid S_0 \right) - \mathbb{E}_P \left(Y_3^{g_0^{0'}, g_1^{0'}, g_2^{0'}} \mid S_0 \right) = \gamma_0^{g^0, g^{0'}}(S_0).$
2. $\mathbb{E}_P \left(Y_3^{A_0, g_1^1, g_2^1} \mid \bar{S}_1 \right) - \mathbb{E}_P \left(Y_3^{A_0, g_1^{1'}, g_2^{1'}} \mid \bar{S}_1 \right) = \gamma_1^{g^1, g^{1'}}(\bar{S}_1).$
3. $\mathbb{E}_P \left(Y_3^{\bar{A}_1, g_2^2} \mid \bar{S}_2 \right) - \mathbb{E}_P \left(Y_3^{\bar{A}_1, g_2^{2'}} \mid \bar{S}_2 \right) = \gamma_2^{g^2, g^{2'}}(\bar{S}_2).$

However, in this paper, we treat the causal contrasts as maps from the distribution space to vector spaces with elements to be functions of the time-varying confounders. To properly define expectations under all distributions of interest, we need to restrict to the distribution space by the following conditions.

Condition 1. (*Absolutely continuous with respect to a probability measure.*) *There exists a common probability measure λ , such that for every $P \in \mathcal{P}$, $P \ll \lambda$, and its Radon-Nikodym derivative $p = dP/d\lambda$ follows that $\int p(o)^2 d\lambda(o) < \infty$, i.e., $p \in L^2(\lambda)$.*

Condition 2. (*Real-valued outcome.*) *Under all $P \in \mathcal{P}$, Y and all elements in the set $\{Y_{K+1}^{\bar{a}_K} : \bar{a}_K \in \{0, 1\}^{K+1}\}$ are real-valued. The distribution P does not impose any addition restrictions on their conditional mean except that $\mathbb{E}_P(Y_{K+1}^{\bar{A}_m, \underline{a}_m} \mid \bar{S}_m) \in L_2(\lambda)$.*

Condition 3. (*Consistency.*) *For every $P \in \mathcal{P}$, under P , if $\bar{A}_K = \bar{a}_K$, then $Y_{K+1}^{\bar{a}_K} = Y$.*

Condition 4. (*Sequential positivity with bounded propensity score.*) *For every $P \in \mathcal{P}$, under P , there exists ϵ_P such that $\epsilon_P < P(A_m = 1 \mid \bar{S}_m) < 1 - \epsilon_P$, for all $m = 0, \dots, K$.*

Condition 5. (*Sequential exchangeability*) *For every $P \in \mathcal{P}$, under P , $Y_{K+1}^{\bar{a}_{m-1}, \underline{g}_m} \perp\!\!\!\perp A_m \mid \bar{S}_m = \bar{s}_m$, for all treatment arms g_m and all $\bar{s}_m \in \mathcal{S}_m$, $m = 0, \dots, K$.*

Conditions 1 and 2 ensure that the conditional expectation of Y and its counterfactual are well-defined without additional constraints. As a counter-example, consider the running example in

Section 1, but now $Y = \xi_0 + \xi_1 L_0 + \xi_2 A_0 + \varepsilon$. This distribution violates Condition 2 because there is a constraint that $\mathbb{E}(Y \mid \bar{S}_1, A_1 = a_1)$ are the same regardless of the value of a_1 . In contrasts, the saturated modeling in the running example satisfies the condition.

Conditions 3 and 5 are the usual consistency and exchangeability assumptions in the sequential trials. Although the bounded score condition in Condition 4 seems stronger than the usual positivity assumption, it is actually widely assumed in causal inference literature especially those that use double robust cross-fitted estimators; e.g., see Theorem 3 in [Kennedy et al. \(2023\)](#). In a word, we restrict our attention to distribution classes that causal inference literature cares about.

2.1 Causal contrasts: maps from distributions to square integrable functions

We now are ready to define causal contrasts properly. Let us set $\Gamma_m = \{\gamma_m : \bar{S}_m \mapsto \mathbb{R}\} \cap L_2(\lambda)$, and define the Trial m causal contrast under the arms $g^m, g^{m,\prime}$ as the map $\tau_m^{g^m, g^{m,\prime}} : \mathcal{P} \longrightarrow \Gamma_m$ given by

$$\tau_m^{g^m, g^{m,\prime}}(P) = \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, g_m^m} \mid \bar{S}_m \right) - \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, g_m^{m,\prime}} \mid \bar{S}_m \right),$$

whose value is determined by the full-data distribution P as well as the two arms $g^m, g^{m,\prime}$. Since we usually only compare one pair of arms in one trial, for notational simplicity, we write $\tau_m^{g^m, g^{m,\prime}}$ as τ_m without ambiguity. Conditions 1 and 2 ensure that the expectations in $\tau_m^{g^m, g^{m,\prime}}(P)$ is finite for all $P \in \mathcal{P}$. Furthermore, for any $\gamma_m \in \Gamma_m$ that is induced by some $P \in \mathcal{P}$, the condition $\gamma_m \in L_2(\lambda)$ and $p, p' \in L_2(\lambda)$ ensures that $\mathbb{E}_{P'} \{\gamma_m(\bar{S}_m)\} < \infty$ because

$$\mathbb{E}_{P'} \{|\gamma_m(\bar{S}_m)|\} = \int |\gamma_m(\bar{s}_m)| p(x) d\lambda(x) \leq \left\{ \int |\gamma_m(\bar{s}_m)|^2 d\lambda(x) \right\}^{1/2} \left\{ \int |p_m(x)|^2 d\lambda(x) \right\}^{1/2},$$

which is less than ∞ . This property is useful in the proof for variation-independence because it alleviates the concern that the causal contrasts induced by a particular distribution may not be integrable under another distribution.

Now, we are going to define the variation-independence property. Define the marginal range of $\tau_m^{g^m, g^{m,\prime}}$ by $\mathcal{T}_m^{g^m, g^{m,\prime}} = \tau_m^{g^m, g^{m,\prime}}(\mathcal{P}) = \{\tau_m^{g^m, g^{m,\prime}}(P) : P \in \mathcal{P}\}$. Define the joint causal contrast

map $\tau^{g^m, g^{m, \prime}} : \mathcal{P} \rightarrow \prod_{m=0}^K \Gamma_m^{g^m, g^{m, \prime}}$ by $\tau(P) = (\tau_0(P), \dots, \tau_K(P))$, and define its joint range by $\mathcal{T}^{g^m, g^{m, \prime}} = \left\{ (\tau_0^{g^0, g^{0, \prime}}(P), \dots, \tau_K^{g^K, g^{K, \prime}}(P)) : P \in \mathcal{P} \right\}$. The causal contrasts $\tau_0^{g^0, g^{0, \prime}}, \dots, \tau_K^{g^K, g^{K, \prime}}$ are jointly variation-independent under $\mathcal{P}$ if and only if $\mathcal{T}^{g^m, g^{m, \prime}} = \prod_{m=0}^K \mathcal{T}_m^{g^m, g^{m, \prime}}$. Equivalently, for every collection of marginal contrast functions $\eta_m \in \mathcal{T}_m^{g^m, g^{m, \prime}}$, $m = 0, \dots, K$, there exists a single law $P_\eta \in \mathcal{P}$ such that $\tau_m^{g^m, g^{m, \prime}}(P_\eta) = \eta_m$ for every $m = 0, \dots, K$.

Note that the range of conditional causal contrasts for $m = 0, \dots, K$ and their joint range depends on the choice of treatment arms $\{g^m, g^{m, \prime}\}_{m=0}^K$. Consequently, the variation-dependence property depends on the choice of the treatment arms. We will show that without constraints on $\{g^m, g^{m, \prime}\}_{m=0}^K$ the causal contrasts can be variation-dependent across trials. For notational simplicity, we will omit the superscripts $g^m, g^{m, \prime}$ in the case without ambiguity.

2.2 Structural nested mean models

Theorem 8.5 in [Robins et al. \(2000\)](#) establishes that additive SNMMs are variation-independent when the conditional means of counterfactual outcomes are unbounded on the real line. To understand this result, it is helpful to examine the causal contrasts modeled by an additive SNMM. In the three-time case, these contrasts are

1. $\mathbb{E}_P \left(Y_3^{a_0, h_1, h_2} - Y_3^{h_0, h_1, h_2} \mid S_0, A_0 = a_0 \right)$, for $a_0 = 0, 1$.
2. $\mathbb{E}_P \left(Y_3^{A_0, a_1, h_2} - Y_3^{A_0, h_1, h_2} \mid \bar{S}_1, A_1 = a_1 \right)$, for $a_1 = 0, 1$.
3. $\mathbb{E}_P \left(Y_3^{A_0, A_1, a_2} - Y_3^{A_0, A_1, h_2} \mid \bar{S}_2, A_2 = a_2 \right)$, for $a_2 = 0, 1$.

Here, h_0, h_1, h_2 are prespecified functions from $\mathcal{S}_0, \bar{\mathcal{S}}_1, \bar{\mathcal{S}}_2$ to $\{0, 1\}$, respectively, and belong to the family of treatment arms. Following convention, we refer to h_0, h_1, h_2 as “treatment regimes,” although they are mathematically equivalent to the generic treatment arms.

Compared with the general causal contrasts in the corresponding trials, the SNMM causal contrasts additionally condition on treatment. We use Trial 0 to illustrate how these contrasts can be simplified to facilitate comparison with the general causal contrasts. In Trial 0, if $a_0 = h_0(s_0)$,

the causal contrast is

$$\mathbb{E}_P \left(Y_3^{h_0, h_1, h_2} - Y_3^{h_0, h_1, h_2} \mid S_0 = s_0, A_0 = h_0(S_0) \right),$$

which is zero by definition. Thus, the nontrivial case is $a_0 = 1 - h_0(s_0)$, which yields the causal contrast

$$\mathbb{E}_P \left(Y_3^{1-h_0, h_1, h_2} - Y_3^{h_0, h_1, h_2} \mid S_0 = s_0, A_0 = 1 - h_0(S_0) \right) = \mathbb{E}_P \left(Y_3^{1-h_0, h_1, h_2} - Y_3^{h_0, h_1, h_2} \mid S_0 = s_0 \right),$$

where the equality holds under the exchangeability assumption. Similarly, we can rewrite the SNMM causal contrasts for Trials 1 and 2 as

$$\mathbb{E}_P \left(Y_3^{A_0, 1-h_1, h_2} - Y_3^{A_0, h_1, h_2} \mid \bar{S}_1 \right), \text{ and, } \mathbb{E}_P \left(Y_3^{A_0, A_1, 1-h_2} - Y_3^{A_0, A_1, h_2} \mid \bar{S}_2 \right)$$

These contrasts are similar to the causal contrasts of interest under the general treatment arms $\{g^m, g^{m'}\}_{m=0}^K$, with two distinctions. First, at the start of each trial, the two arms in an SNMM contrast assign different treatment values. For example, at time 1, one arm follows $h_1(\bar{S}_1)$, and the other follows $1 - h_1(\bar{S}_1)$. Second, after the trial begins, the two arms in an SNMM contrast follow the same regime, whereas the general treatment arms can follow different functions. For example, at calendar time 2, the general treatment arms allow the treatment values to be $g_2^0(\bar{S}_2), g_2^{0'}(\bar{S}_2)$ in Trial 0, $g_2^1(\bar{S}_2), g_2^{1'}(\bar{S}_2)$ in Trial 1, and $g_2^2(\bar{S}_2), g_2^{2'}(\bar{S}_2)$ in Trial 2, whereas the SNMM restricts the treatment values to $A_2, h_2(\bar{S}_2)$ across all three trials. In this paper, we show that the first restriction suffices to ensure variation-independence of the causal contrasts across trials. The second restriction enables the construction of doubly robust g -estimators but is not required for variation-independence.

3 VARIATION-INDEPENDENT CAUSAL CONTRASTS

In this section, we establish a necessary and sufficient condition for causal contrasts under static regimes to be variation-independent across trials. The results for dynamic regimes are more involved, and we establish only a sufficient condition, which is presented in the appendix. By definition, a dynamic regime at time m is a function $\bar{\mathcal{S}}_m \mapsto \{0, 1\}$. Because this function may depend on time-varying covariates affected by prior treatment, the treatment assigned at a given time may be random and cannot generally be determined before the trial begins. A static regime is a special case in which the treatment rule maps only $\bar{\mathcal{A}}_m \mapsto \{0, 1\}$, so treatment assignments depend only on prior treatment history. Consequently, all treatment values can be determined before the trial begins. For example, in the three-time case, one arm in Trial 0 can be specified as $\{1, g_1^0(A_0), g_2^0(A_1)\}$, with $g_1^0(A_0) = 1 - A_0$ and $g_2^0(A_1) = 1 - A_1$. The treatment sequence under this regime is therefore $(1, 0, 1)$, which is known before the trial begins. See Fine Point 19.1 of [Miguel et al. \(2023\)](#) and [Guo et al. \(2021\)](#) for detailed definitions.

In a trial, researchers seek to compare arms that yield meaningful causal contrasts. For example, it is uninformative to compare two arms in Trial 0 specified by $g_0^0(s_0) = 1, g_1^0(a_0) = 1 - a_0, g_2^0(\bar{a}_1) = 1 - a_0$ and $g_0^{0'}(s_0) = 1, g_1^{0'}(a_0) = \mathbb{I}(a_0 > 10), g_2^{0'}(\bar{a}_1) = a_1$. Although g^0 and $g^{0'}$ have different functional forms, they yield the same treatment sequence, $(1, 0, 0)$. Their causal contrast is therefore identically zero and, by definition, variation-independent of the causal contrasts in all other trials. To exclude this trivial case, we impose the following condition.

Condition 6 (Distinct static treatment arms). *For each $m = 0, \dots, K$ and every $\bar{a}_{m-1} \in \bar{\mathcal{A}}_{m-1}$, the two static treatment arms in Trial m are distinct; that is,*

$$\underline{g}_m^m(\bar{a}_{m-1}) \neq \underline{g}_m^{m'}(\bar{a}_{m-1}).$$

Equivalently, for every $m = 0, \dots, K$ and every $\bar{a}_{m-1} \in \bar{\mathcal{A}}_{m-1}$, there exists some $k \in \{m, \dots, K\}$ such that $a_k^m(\bar{a}_{m-1}) \neq a_k^{m'}(\bar{a}_{m-1})$.

Under Condition 6, for each prior treatment history $\bar{a}_{m-1}$, the two arms assign different treatment values at least once between times m and K .

Theorem 1 (A necessary and sufficient condition for variation independence under static arms). *Let $\mathcal{P}$ be the class of all full-data laws satisfying Conditions 1, 2, 3, 4, and 5. Fix $m \in \{0, \dots, K\}$ and a collection of static treatment arms $\{g^j, g^{j'}\}_{j=m}^K$ for Trials $m, \dots, K$. Suppose that the corresponding causal contrasts from Trials $m+1$ are jointly variation-independent under $\mathcal{P}$ and that the two arms in Trial m satisfy the distinctness requirement in Condition 6.*

The causal contrasts induced by treatment arms $\{g^j, g^{j'}\}_{j=m}^K$ are jointly variation-independent under $\mathcal{P}$ if and only if

$$g_m^m(\bar{a}_{m-1}) \neq g_m^{m'}(\bar{a}_{m-1})$$

for every $\bar{a}_{m-1} \in \{0, 1\}^m$. For $m = K$, the assumption concerning subsequent trials is vacuous.

Theorem 1 can be understood by working backwards from Trial K . At time K , only one treatment value remains to be determined, so a meaningful contrast must compare treatment with no treatment, i.e., a 1-0 comparison in $\mathbb{E}\left(Y_{K+1}^{\bar{A}_{K-1}, g_K^K} - Y_{K+1}^{\bar{A}_{K-1}, 1-g_K^K} \mid \bar{S}_K\right)$. This contrast satisfies the condition in Theorem 1 by setting $g_K^{K'}(\bar{S}_K) = 1 - g_K^K(\bar{S}_K)$. Next, to construct a meaningful causal contrast in Trial $K-1$ that is variation-independent of the Trial K contrast under $\mathcal{P}$, the two treatment arms must satisfy $g_{K-1}^{K-1}(\bar{a}_{K-1}) \neq g_{K-1}^{K-1'}(\bar{a}_{K-1})$. The contrasts under treatment arms $\{g^j, g^{j'}\}_{j=K-1}^K$ are then jointly variation-independent. Continuing backwards in this way yields jointly variation-independent causal contrasts for Trials 0 through K .

To see why the condition is necessary, consider a counterexample with three time points. In Trial 1, let $g_1^1(a_0) = g_1^{1'}(a_0) = 0$, $g_2^1(\bar{a}_1) = 1$, and $g_2^{1'}(\bar{a}_1) = 0$. In Trial 2, let $g_2^2(\bar{a}_1) = 1$ and $g_2^{2'}(\bar{a}_1) = 0$. By conditional exchangeability and iterated expectations, the Trial 1 causal contrast

satisfies

$$\begin{aligned}
& \mathbb{E}_P \left(Y_3^{A_0,0,1} - Y_3^{A_0,0,0} \mid \bar{S}_1 = \bar{s}_1 \right) \\
&= \mathbb{E}_P \left(Y_3^{A_0,0,1} - Y_3^{A_0,0,0} \mid \bar{S}_1 = \bar{s}_1, A_1 = 0 \right) \\
&= \mathbb{E}_P \left(Y_3^{A_0,A_1,1} - Y_3^{A_0,A_1,0} \mid \bar{S}_1 = \bar{s}_1, A_1 = 0 \right) \\
&= \int \mathbb{E}_P \left(Y_3^{A_0,A_1,1} - Y_3^{A_0,A_1,0} \mid \bar{S}_1 = \bar{s}_1, A_1 = 0, L_2 = l_2 \right) dP(l_2 \mid \bar{S}_1 = \bar{s}_1, A_1 = 0).
\end{aligned}$$

The conditional expectation inside the integral is the Trial 2 causal contrast evaluated at the corresponding history $\bar{s}_2$. Thus, the Trial 1 contrast is an average of the Trial 2 contrast over the conditional distribution of L_2 . In particular, under any law $P \in \mathcal{P}$ for which the Trial 2 contrast is zero almost surely, the Trial 1 contrast must also be zero almost surely. Therefore, for these treatment arms, the two contrasts cannot vary freely and are variation-dependent.

The counterexample shows how variation dependence can arise when two trial arms assign the same treatment at the start: the contrast between them may be an average of a later trial's contrast. To avoid this constraint, consider arms that assign different treatments at the start. For example, let $g_1^1(a_0) = 0$, $g_1^{1'}(a_0) = 1$, and $g_2^1(\bar{a}_1) = g_2^{1'}(\bar{a}_1) = 0$. As illustrated in Figure 2, these arms diverge at time 1 and both assign treatment 0 at time 2. To establish variation independence, take any two laws $P, P' \in \mathcal{P}$ and let their respective Trial 2 and Trial 1 contrasts be

$$\mathbb{E}_P \left(Y^{\bar{A}_1,1} - Y^{\bar{A}_1,0} \mid \bar{S}_2 \right) = \gamma_2(\bar{S}_2), \quad \mathbb{E}_{P'} \left(Y^{A_0,0,0} - Y^{A_0,1,0} \mid \bar{S}_1 \right) = \gamma_1(\bar{S}_1).$$

To prove the variation-independence, we are looking for a single law $P'' \in \mathcal{P}$ that yields both contrasts. Such a distribution law can be constructed by first choosing an arbitrary joint distribution of $(L_0, A_0, L_1, A_1, L_2, A_2)$ satisfying the required positivity and integrability conditions. Then we specify the conditional outcome distribution as

$$\mathbb{E}_{P''}(Y \mid \bar{S}_2, A_2) = A_2 \gamma_2(\bar{S}_2) - A_1 \gamma_1(\bar{S}_1).$$

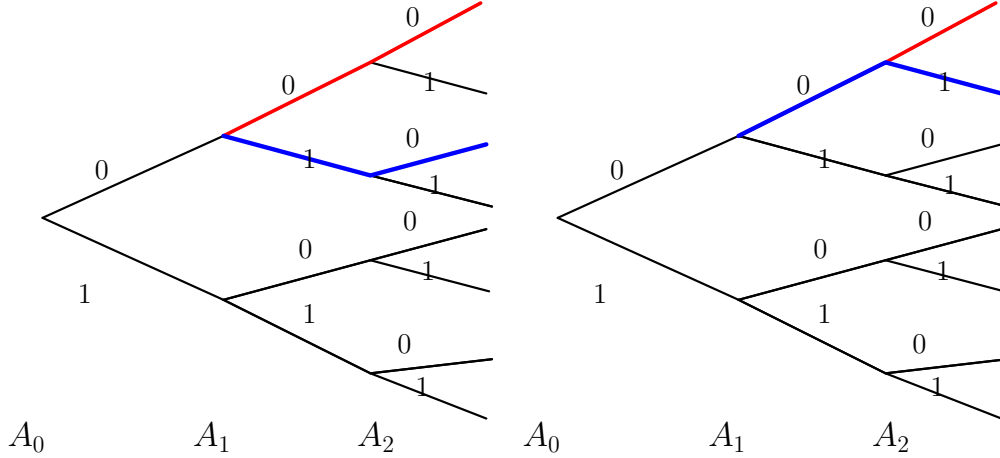


Figure 2: Causal trees illustrating variation independence. The left panel shows the causal contrast $\mathbb{E}(Y^{0,0,0} - Y^{0,1,0} \mid \bar{S}_1)$, and the right panel shows the contrast $\mathbb{E}(Y^{0,0,0} - Y^{0,0,1} \mid \bar{S}_1)$.

The resulted distribution fulfills the requirement. One can use the g -formula to verify that

$$\mathbb{E}_{P''} \left(Y^{\bar{A}_1,1} - Y^{\bar{A}_1,0} \mid \bar{S}_2 \right) = \gamma_2(\bar{S}_2), \quad \mathbb{E}_{P''} \left(Y^{A_0,0,0} - Y^{A_0,1,0} \mid \bar{S}_1 \right) = \gamma_1(\bar{S}_1).$$

Thus, any attainable Trial 1 contrast can be combined with any attainable Trial 2 contrast under a single law in $\mathcal{P}$, establishing their variation-independence. This construction can be generalized to the argument used to prove sufficiency in Theorem 1.

4 HA-MSM FAILS BUT SNMM MEETING THEOREM 1

As illustrated in Section 1, the parameters of an HA-MSM may be variation-dependent. Theorem 1 helps explain the source of this variation-dependence. In the three-time case, consider an HA-MSM under P specified by

1. $\mathbb{E}_P(Y_3^{a_0,a_1,a_2} \mid S_0 = s_0) = \beta_0(a_0, a_1, a_2, s_0; \alpha_0)$ in Trial 0,
2. $\mathbb{E}_P(Y_3^{A_0,a_1,a_2} \mid \bar{S}_1 = \bar{s}_1) = \beta_1(a_0, a_1, a_2, \bar{s}_1; \alpha_1)$ in Trial 1,
3. $\mathbb{E}_P(Y_3^{A_0,A_1,a_2} \mid \bar{S}_2 = \bar{s}_2) = \beta_2(a_0, a_1, a_2, \bar{s}_2; \alpha_2)$ in Trial 2,

where $\beta_0, \beta_1, \beta_2$ are parametric models indexed by $\alpha_0, \alpha_1, \alpha_2$, respectively.

These conditional means imply contrasts that are variation-dependent, regardless of the chosen functional forms. In particular, the Trial 1 model implies

$$\mathbb{E}_P \left(Y_3^{A_0,0,1} - Y_3^{A_0,0,0} \mid \bar{S}_1 = \bar{s}_1 \right) = \beta_1(a_0, 0, 1, \bar{s}_1; \alpha_1) - \beta_1(a_0, 0, 0, \bar{s}_1; \alpha_1).$$

This is the contrast in the counterexample above, which is variation-dependent on the Trial 2 contrast

$$\mathbb{E}_P \left(Y_3^{A_0,A_1,1} - Y_3^{A_0,A_1,0} \mid \bar{S}_2 = \bar{s}_2 \right) = \beta_2(a_0, a_1, 1, \bar{s}_2; \alpha_2) - \beta_2(a_0, a_1, 0, \bar{s}_2; \alpha_2).$$

Thus, the variation-independence issue arises because HA-MSMs attempt to model more causal contrasts than can vary independently across sequential trials.

As a comparison, SNMM models less causal contrasts, and restrict to those starting with different treatment values. Actually, the following proposition shows that SNMMs always model a 0-1 comparison at the beginning of each trial

Proposition 1. *For $m = 0, \dots, K$, $P \in \mathcal{P}$, under Assumptions 3, 4, and 5,*

$$\mathbb{E}_P \left(Y_{K+1}^{\bar{A}_m, \underline{h}_{m+1}} - Y_{K+1}^{\bar{A}_{m-1}, \underline{h}_m} \mid \bar{S}_m = \bar{s}_m, A_m = a_m \right) = \{a_m - h_m(\bar{s}_m)\} \gamma_m(\bar{s}_m),$$

if and only if

$$\mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}} - Y_{K+1}^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m \right) = \gamma_m(\bar{s}_m),$$

for all $\bar{s}_m \in \bar{\mathcal{S}}_m$ and $a_m \in \{0, 1\}$.

Thus, SNMM always satisfy the condition in Theorem 1 and are variation-independent.

5 CHARACTERIZING EQUIVALENCE TO SNMMS

Theorem A.1 shows that the first distinguishing property of SNMMs ensures that their causal contrasts are variation-independent. We now ask what further distinguishes SNMM contrasts from general variation-independent causal contrasts. To establish this connection, we first express the treatment arms represented by g 's in terms of treatment regimes represented by h 's.

Proposition 2 (Reparameterization by a common regime). *Fix $m \in \{0, \dots, K\}$, and let $\underline{g}_m^m = (g_m^m, \dots, g_K^m)$ and $\underline{g}_m^{m'} = (g_m^{m'}, \dots, g_K^{m'})$ denote the two treatment arms of Trial m . Suppose that $A_j \in \{0, 1\}$ for $j = m, \dots, K$ and that $g_m^m(\bar{s}_m) \neq g_m^{m'}(\bar{s}_m)$ for every $\bar{s}_m \in \bar{\mathcal{S}}_m$. Then there exists a common regime $\underline{h}_{m+1}^m = (h_{m+1}^m, \dots, h_K^m)$ such that, for all $P \in \mathcal{P}$ and all $\bar{s}_m \in \bar{\mathcal{S}}_m$,*

$$\begin{aligned} & \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, g_m^m, \underline{g}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, g_m^{m'}, \underline{g}_{m+1}^{m'}} \mid \bar{\mathcal{S}}_m = \bar{s}_m \right) \\ &= \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, g_m^m, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, 1 - g_m^m, \underline{h}_{m+1}^m} \mid \bar{\mathcal{S}}_m = \bar{s}_m \right). \end{aligned}$$

For each $j = m + 1, \dots, K$, we can construct a $h_j^m : \bar{\mathcal{S}}_j \rightarrow \{0, 1\}$ by

$$h_j^m(\bar{s}_j) = g_j^m(\bar{s}_j) \mathbb{I} \{a_m = g_m^m(\bar{s}_m)\} + g_j^{m'}(\bar{s}_j) \mathbb{I} \{a_m = 1 - g_m^m(\bar{s}_m)\}.$$

When $m = K$, $\underline{h}_{K+1}^K$ is understood to be the empty regime.

Proposition 2 shows that, under the stated conditions, the two treatment arms can be represented using a common regime $\underline{h}_{m+1}^m$ after time m . This regime reproduces the treatment assignments specified by each original arm by conditioning on the treatment received at time m . The two arms therefore differ only in their initial treatment rules, g_m^m and $g_m^{m'} = 1 - g_m^m$. The next proposition expresses this contrast as a comparison of treatment 1 versus treatment 0 at time m , followed by the common regime $\underline{h}_{m+1}^m$.

Proposition 3. *For causal contrasts in Proposition 2,*

$$\mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, g_m^m, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, 1-g_m^m, \underline{h}_{m+1}^m} \mid \bar{S}_m = \bar{s}_m \right) = g_m^m(\bar{s}_m) \gamma_m^m(P)(\bar{s}_m) - \{1 - g_m^m(\bar{s}_m)\} \gamma_m^m(P)(\bar{s}_m)$$

if and only if

$$\mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}^m} \mid \bar{S}_m = \bar{s}_m \right) = \gamma_m(\bar{s}_m),$$

for all $P \in \mathcal{P}$ and $\bar{s}_m \in \bar{\mathcal{S}}_m$.

Comparing these results with Proposition 1 reveals a key distinction: the regime functions here carry an additional superscript m , allowing the subsequent treatment regime to vary across trials. In contrast, an SNMM uses a common treatment regime across trials. Requiring the trial-specific regime functions to agree almost surely under P at each treatment time shared by the trials is therefore sufficient for the causal contrasts to be equivalent to those generated by an SNMM. However, this condition is stronger than necessary. The following theorem establishes such a necessary and sufficient condition for static arms.

Theorem 2. *When $\underline{h}_{m+1}^m$ are static treatment arms, the variation-independent causal contrasts*

$$\mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}^m} \mid \bar{S}_m = \bar{s}_m \right) = \gamma_m(\bar{s}_m),$$

with $0 \leq m \leq K$ form an SNMM for all $P \in \mathcal{P}$ satisfying Conditions 1, 2, 3, 4, 5, 6, if and only if

$$h_k^m \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \} = h_k^j \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \},$$

for $0 \leq m < j \leq k-1 \leq K-1$ and all $\bar{a}_m \in \bar{\mathcal{A}}_m$.

The equivalence in Theorem 1 is required to hold for every $P \in \mathcal{P}$ satisfying the stated conditions, rather than for a single law P . This requirement rules out accidental equivalences under particular laws. For example, under a law in which all treatment regimes yield the same conditional

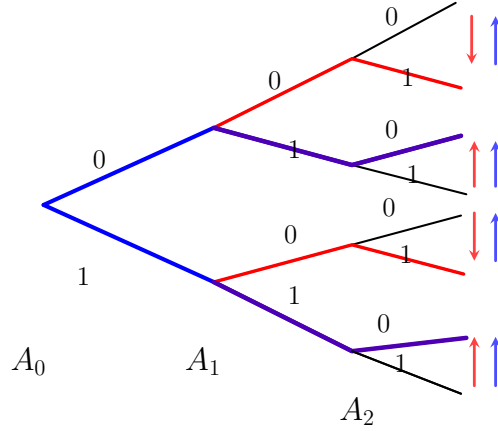


Figure 3: Causal trees illustrating SNMM-equivalent causal contrasts

mean outcome, all causal contrasts are zero. In that case, arbitrary treatment arms yield the same contrasts as an SNMM, which are all zero.

The condition in Theorem 2 requires the regimes in Trials m and j to assign the same treatment at time k along treatment histories that follow the regime of Trial m from time $m + 1$ through time $k - 1$. They need not agree for all treatment histories. To illustrate this distinction, consider Figure 3, where we compare the following causal contrasts:

1. $\mathbb{E}(Y_3^{0,1,0} - Y_3^{1,1,0} \mid S_0)$ at Trial 0.
2. $\mathbb{E}(Y_3^{A_0,0,1} - Y_3^{A_0,1,0} \mid \bar{S}_1, A_0 = a_0)$ for $a_0 = 0, 1$ at Trial 1.
3. $\mathbb{E}(Y_3^{A_0,A_1,1} - Y_3^{A_0,A_1,0} \mid \bar{S}_2, A_0 = a_0, A_1 = a_1)$ for $a_0, a_1 \in \{0, 1\}$ at Trial 2.

The contrasts in Trial 2 already compare treatment 1 with treatment 0 at the final treatment time, so no subsequent regime needs to be specified. Applying the reparameterization in Proposition 2, we can rewrite the contrasts in Trials 0 and 1 as

1. Trial 0: $\mathbb{E}(Y^{0,h_1^0,h_2^0} - Y^{1,h_1^0,h_2^0} \mid S_0)$, where $h_1^0(a_0) = 1$ and $h_2^0(a_0, a_1) = 0$.
2. Trial 1: $\mathbb{E}(Y^{A_0,0,h_2^1} - Y^{A_0,1,h_2^1} \mid \bar{S}_1, A_0 = a_0)$, where $h_2^1(a_0, a_1) = 1 - a_1$, for each $a_0 \in \{0, 1\}$.

For each $a_0 \in \{0, 1\}$, $h_2^0\{a_0, h_1^0(a_0)\} = h_2^1\{a_0, h_1^0(a_0)\} = 0$. Thus, although the two regimes differ when $a_1 = 0$, they agree along every treatment history that follows the regime of Trial 0. The

condition in Theorem 2 therefore holds. After reversing the signs of the contrasts in Trials 0 and 1, the three contrasts correspond to the blip functions of an SNMM:

$$\mathbb{E}(Y^{1,h_1,h_2} - Y^{0,h_1,h_2} \mid S_0), \mathbb{E}(Y^{A_0,1,h_2} - Y^{A_0,0,h_2} \mid \bar{S}_1), \mathbb{E}(Y^{A_0,A_1,1} - Y^{A_0,A_1,0} \mid \bar{S}_2),$$

with $h_1(a_0) = 1$ and $h_2(a_0, a_1) = 1 - a_1$.

The condition $h_2^0\{a_0, h_1^0(a_0)\} = h_2^1\{a_0, h_1^0(a_0)\}$ can also be assessed visually by comparing the blip directions. In Figure 3, blue and red arrows indicate the blip directions in Trials 0 and 1, respectively; blue and red branches indicate the treatment trajectory that people would follow under the assigned treatment arms in Trials 0 and 1. In Trial 0, the regime assigns $A_2 = 0$ for every history (A_0, A_1) , so all blue arrows point toward treatment 0. In Trial 1, the regime assigns $A_2 = 1 - A_1$, so the red arrows point toward treatment 0 when $A_1 = 1$ and toward treatment 1 when $A_1 = 0$. Thus, the two trials share the same blip directions on the branches $(A_0, A_1) = (0, 1)$ and $(1, 1)$, but have different directions on the branches $(A_0, A_1) = (0, 0)$ and $(1, 0)$.

The condition in Theorem 2 requires agreement only along histories satisfying $A_1 = h_1^0(A_0) = 1$, namely, the branches $(A_0, A_1) = (0, 1)$ and $(1, 1)$. Therefore, although the blip directions differ on the other branches, the condition still holds. After reversing the sign before the contrasts as, they correspond to the blip functions of an SNMM.

Now consider the counterexample in Figure 4. We replace the contrast in Trial 0 with $\mathbb{E}(Y^{0,1,1} - Y^{1,1,1} \mid S_0)$, while leaving the contrasts in Trials 1 and 2 unchanged. The modified regime in Trial 0 is given by $h_1^0(a_0) = 1$ and $h_2^0(a_0, a_1) = 1$. Along histories that follow this regime, the treatment assignments at time 2 now differ between Trials 0 and 1: for each $a_0 \in \{0, 1\}$, $h_2^0\{a_0, h_1^0(a_0)\} = 1$, and $h_2^1\{a_0, h_1^0(a_0)\} = 0$. Thus, the condition in Theorem 2 fails. Visually, this failure is reflected in the opposite red and blue arrows on the branches $(A_0, A_1) = (0, 1)$ and $(1, 1)$, which follow the regime of Trial 0.

The Venn diagram in Figure 5 summarizes Theorems 1 and 2. Sequential trials admit a broad class of causal models, and Theorem 1 gives conditions under which their causal contrasts are

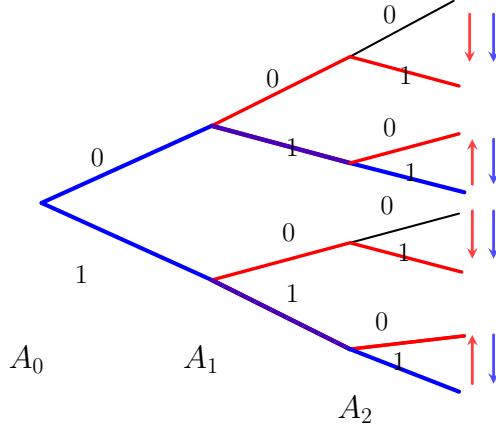


Figure 4: Causal trees illustrating causal contrasts that are not SNMM-equivalent

variation-independent. HA-MSMs need not satisfy these conditions and can exhibit variation dependence. Within the class of variation-independent causal contrasts, Theorem 2 characterizes the subset equivalent to those generated by SNMMs, for which established g -estimation methods are available.

6 WHY G -ESTIMATION FAILS

We now turn to variation-independent causal contrasts outside the SNMM-equivalent subset, represented by the region between the two circles. Variation-independence allows these contrasts to be modeled without compatibility constraints, but they do not admit an SNMM representation. Whether this limitation poses a problem depends on the estimation method. For example, inverse probability weighting (IPW) can still be used to estimate these contrasts without concerns about overparameterization. However, the standard g -estimations, which are usually more efficient than IPW, are no longer available. We therefore examine why g -estimation argument fails for these contrasts.

To illustrate the failure reason, consider the three-time example depicted in Figures 3 and 4,

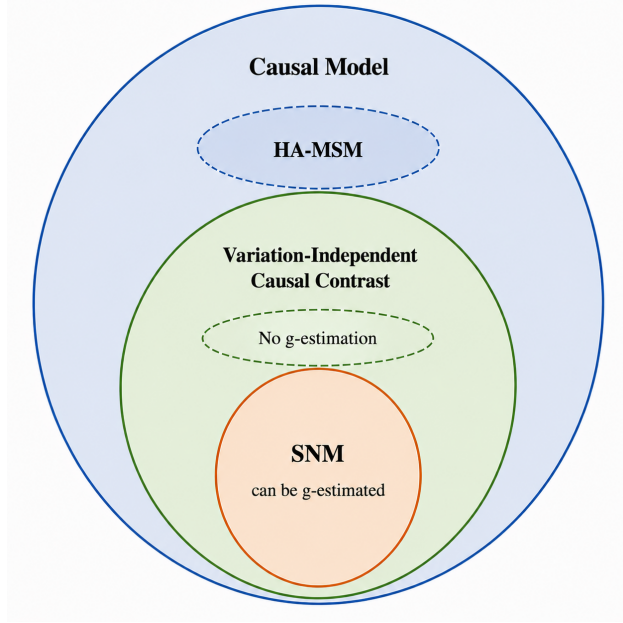


Figure 5: Caption

with the blip functions to be,

$$\begin{aligned}\mathbb{E}(Y_3^{A_0, A_1, 1} - Y_3^{A_0, A_1, 0} \mid \bar{S}_2) &= \gamma_2(\bar{S}_2), \\ \mathbb{E}(Y_3^{A_0, 1, h_2^1} - Y_3^{A_0, 0, h_2^1} \mid \bar{S}_1) &= \gamma_1(\bar{S}_1), \\ \mathbb{E}(Y_3^{1, h_1^0, h_2^0} - Y_3^{0, h_1^0, h_2^0} \mid S_0) &= \gamma_0(S_0).\end{aligned}$$

As in the literature for the g -estimation, one needs to specify the following

$$\begin{aligned}\pi_0(S_0) &= \Pr(A_0 = 1 \mid S_0), \quad \pi_1(\bar{S}_1) = \Pr(A_1 = 1 \mid \bar{S}_1), \quad \pi_2(\bar{S}_2) = \Pr(A_2 = 1 \mid \bar{S}_2), \\ \mu_2^2(\bar{S}_2) &= \mathbb{E}(Y_3 \mid \bar{S}_2, A_2 = 0), \\ \mu_1^1(\bar{S}_1) &= \mathbb{E} \{ \mathbb{E}(Y_3 \mid \bar{S}_2, A_2 = h_2^1) \mid \bar{S}_1, A_1 = 0 \}, \\ \mu_0^0(S_0) &= \mathbb{E} [\mathbb{E} \{ \mathbb{E}(Y_3 \mid \bar{S}_2, A_2 = h_2^0) \mid S_1, A_1 = h_1^0 \} \mid S_0, A_0 = 0],\end{aligned}$$

where for $j = 0, 1, 2$. For $0 \leq j \leq m \leq 2$, π_m denotes the propensity score at time j , and μ_m^m denotes a recursive outcome regression associated with Trial m and models the outcome mean if the people had no treatment at time m and follows treatment regime $\underline{h}_{m+1}^m$ from time $m + 1$. Under

the identification assumptions, if the causal contrasts admit an SNMM representation, as in Figure 3, then g -estimation yields the following unbiased estimating equations:

$$\mathbb{P} \left[\{A_2 - \pi_2(\bar{S}_2)\} \{Y_3 - \gamma_2(\bar{S}_2)A_2 - \mu_2^2(\bar{S}_2)\} \right] = 0, \quad (2)$$

$$\mathbb{P} \left[\{A_1 - \pi_1(\bar{S}_1)\} \{Y_3 - \gamma_2(\bar{S}_2)(A_2 - h_2^1) - \gamma_1(\bar{S}_1)A_1 - \mu_1^1(\bar{S}_1)\} \right] = 0, \quad (3)$$

$$\mathbb{P} \left[\{A_0 - \pi_0(S_0)\} \{Y_3 - \gamma_2(\bar{S}_2)(A_2 - h_2^1) - \gamma_1(\bar{S}_1)(A_1 - h_1^0) - \gamma_0(S_0)A_0 - \mu_0^0(S_0)\} \right] = 0. \quad (4)$$

When the causal contrasts admit an SNMM or its equivalence, $h_2^1 = h_2^0$, and replacing h_2^1 with h_2^0 in Equation 4 yields the usual g -estimation equation. However, when the causal contrasts do not admit an SNMM representation, as in Figure 4, Equation 4 needs not hold, potentially leading to biased estimation of $\gamma_0(S_0)$. A sufficient condition for this estimating equation to be unbiased is that, at the true parameter values,

$$\mathbb{E}_P \left[Y_3 - \gamma_2(\bar{S}_2; \psi_2)(A_2 - h_2^1) - \gamma_1(\bar{S}_1; \psi_1)(A_1 - h_1^0) \mid \bar{S}_1 \right] = \mathbb{E}_P \left(Y_3^{A_0, h_1^0, h_2^0} \mid \bar{S}_1 \right)$$

holds P -almost surely for every $P \in \mathcal{P}$. However, in the case depicted in Figure 4,

$$\begin{aligned} & \mathbb{E}_P \left\{ Y_3 - \gamma_2(\bar{S}_2; \psi_2)(A_2 - h_2^1) - \gamma_1(\bar{S}_1; \psi_1)(A_1 - h_1^0) \mid \bar{S}_1 \right\} \\ &= \mathbb{E}_P \left[(1 - A_1) \mathbb{E}_P \left\{ Y_3 - \gamma_2(\bar{S}_2; \psi_2)(A_2 - h_2^1) \mid \bar{S}_2, A_2 \right\} \right. \\ & \quad \left. + A_1 \mathbb{E}_P \left\{ Y_3 - \gamma_2(\bar{S}_2; \psi_2)(A_2 - h_2^1) \mid \bar{S}_2, A_2 \right\} - \gamma_1(\bar{S}_1; \psi_1)(A_1 - h_1^0) \mid \bar{S}_1 \right] \\ &= \mathbb{E}_P \left[(1 - A_1) \mathbb{E}_P \left\{ Y_3^{A_0, 0, h_2^0} \mid \bar{S}_2 \right\} + A_1 \mathbb{E}_P \left\{ Y_3^{A_0, 1, h_2^1} \mid \bar{S}_2 \right\} - \gamma_1(\bar{S}_1; \psi_1)(A_1 - h_1^0) \mid \bar{S}_1 \right]. \end{aligned}$$

Thus, the usual blip-down construction targets a different treatment regime from the one defining the Trial 0 contrast. The required conditional mean identity holds only if one of the two conditions holds; 1) conditional on $\bar{S}_1$ $A_1 = 0$, P -a.e. ; or 2) $\mathbb{E}_P \left\{ Y_3^{A_0, 0, h_2^0} \mid \bar{S}_2 \right\} = \mathbb{E}_P \left\{ Y_3^{A_0, 1, h_2^1} \mid \bar{S}_2 \right\}$ P -a.e. for all $P \in \mathcal{P}$. However, since we assume positivity and consider an unrestricted family $\mathcal{P}$,

the above two conditions fail, and there is no guarantee that Equation 4 holds.

Since one only needs to construct $\mathbb{E}_P(Y_3^{A_0, h_1^0, h_2^0} \mid \bar{S}_1)$ from the observed data to obtain an unbiased estimating equation for γ_0 , one could additionally specify the following models:

$$\mu_2^0(\bar{S}_2) = \mathbb{E}(Y_3 \mid \bar{S}_2, A_2 = h_2^0), \mu_1^0(\bar{S}_1) = \mathbb{E}(\mu_2^0 \mid \bar{S}_1, A_1 = h_1^0).$$

However, $\mu_2^0(\bar{S}_2)$ is variation-dependent on $\{\gamma_2(\bar{S}_2), \mu_2^2(\bar{S}_2)\}$ because

$$\mu_2^0(\bar{S}_2) = \mathbb{E}_P(Y_3^{A_0, A_1, h_2^0} \mid \bar{S}_2) = \gamma_2(\bar{S}_2)h_2^0(\bar{S}_2) + \mathbb{E}_P(Y_3^{A_0, A_1, 0} \mid \bar{S}_2) = \gamma_2(\bar{S}_2)h_2^0(\bar{S}_2) + \mu_2^2(\bar{S}_2).$$

Introducing an overparameterized nuisance model solely to perform g -estimation is undesirable. For g -estimation to retain its double robustness, all three functions in $\{\mu_2^0(\bar{S}_2), \gamma_2(\bar{S}_2), \mu_2^2(\bar{S}_2)\}$ must be correctly specified, which is harder because of their variation-dependence. Thus, under this approach, the claimed “double robustness” of the g -estimation estimators no longer holds.

Instead of relying on the recursive regressions above, one could specify the following additional models:

$$\mathbb{E}(Y_3^{A_0, 1, h_2^0} - Y_3^{A_0, 0, h_2^0} \mid \bar{S}_1) = \gamma_1^*(\bar{S}_1; \psi_1^*), \mu_1^0(\bar{S}_1) = \mathbb{E}[Y_3 - \gamma_2(\bar{S}_2; \psi_2)h_2^0(\bar{S}_2) - \gamma_1^*(\bar{S}_1; \psi_1^*)h_1^0(\bar{S}_1) \mid \bar{S}_1],$$

where the asterisk denotes a new blip function for Trial 1. However, this approach would also introduce overparameterization. Given $\gamma_1(\bar{S}_1; \psi_1)$ and $\mu_1^1(\bar{S}_1)$, one can recover the conditional mean

$$\mathbb{E}(Y_3^{A_0, A_1, h_2^1} \mid \bar{S}_1, A_1) = \gamma_1(\bar{S}_1; \psi_1)A_1 + \mu_1^1(\bar{S}_1).$$

In particular,

$$\begin{aligned} \mathbb{E}(Y_3^{A_0, A_1, h_2^1} \mid \bar{S}_1, A_1) &= \gamma_1(\bar{S}_1; \psi_1)A_1 + \mu_1^1(\bar{S}_1) \\ &= \int \{\gamma_2(\bar{S}_2; \psi_2)h_2^1(\bar{S}_2) + \mu_2^2(\bar{S}_2)\} dP(l_2 \mid A_1, \bar{S}_1), \end{aligned}$$

whereas the additional specification implies

$$\begin{aligned}\mathbb{E}(Y_3^{A_0, A_1, h_2^0} \mid \bar{S}_1, A_1) &= \gamma_1^*(\bar{S}_1; \psi_1^*) A_1 + \mu_1^0(\bar{S}_1) \\ &= \int \{ \gamma_2(\bar{S}_2; \psi_2) h_2^0(\bar{S}_2) + \mu_2^2(\bar{S}_2) \} dP(l_2 \mid A_1, \bar{S}_1).\end{aligned}$$

Thus, conditional on $(\bar{S}_1, A_1)$, both means are obtained by integrating with respect to the same conditional law, $P(l_2 \mid A_1, \bar{S}_1)$. Once γ_2 , μ_2^2 , h_2^0 , and h_2^1 are fixed, these conditional means cannot, in general, vary arbitrarily and independently. Consequently, $\{\gamma_1(\bar{S}_1; \psi_1), \mu_1^1(\bar{S}_1)\}$ and $\{\gamma_1^*(\bar{S}_1; \psi_1^*), \mu_1^0(\bar{S}_1)\}$ are subject to compatibility restrictions induced by the common conditional law of L_2 . These compatibility restrictions prevent the blip functions and nuisance functions in this construction from being specified independently.

To conclude, when the blip functions do not admit an SNMM, there is no variation-independent parameterization for the blip functions and all nuisance parameters in g-estimation. Consequently, g-estimation fails.

7 BIAS-CORRECTED g -ESTIMATION

Here, we give the results for a bias-corrected g -estimation and derives the influence function for the variation-independent causal contrasts under arbitrary arms. To illustrate the idea, consider the three-time example. From now on, we will work with a P satisfying Conditions 3, 4, and 5 as those in the usual causal inference literature. We will use $\mathbb{E}$ to denote $\mathbb{E}_P$.

Theorem 3. *Under P satisfying Conditions 3, 4, and 5, it follows that*

$$\begin{aligned}\mathbb{E} \left[\{Y - A_2 \gamma_2(\bar{S}_2)\} \mid \bar{S}_2, A_2 \right] &= \mathbb{E} \left(Y^{A_0, A_1, 0} \mid \bar{S}_2, A_2 \right), \\ \mathbb{E} \left[\{Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - A_1 \gamma_1(\bar{S}_1)\} \mid \bar{S}_1, A_1 \right] &= \mathbb{E} \left\{ Y^{A_1, 0, h_2^1} \mid \bar{S}_1, A_1 \right\}, \\ \mathbb{E} \left[\{Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) - A_0 \gamma_0(S_0) + R \Delta \gamma_2(\bar{S}_2)\} \mid S_0, A_0 \right] &= \mathbb{E} \left(Y^{0, h_1^0, h_2^0} \mid \bar{S}_1, A_0 \right),\end{aligned}$$

where

$$R(A_1, \bar{S}_1) = \frac{\mathbb{I}\{A_1 = h_1^0(\bar{S}_1)\}}{P\{A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1\}}, \quad \Delta(\bar{S}_2) = h_2^0(\bar{S}_2) - h_2^1(\bar{S}_2).$$

The third equation in Theorem 3 is the new interesting result we would like to emphasize. It can be understood as a bias correction for the incorrect blip,

$$Y - (A_2 - h_2^1)\gamma_2(\bar{S}_2) - (A_1 - h_1^0)\gamma_1(\bar{S}_1) - A_0\gamma_0(S_0)$$

whose conditional expectation is $\mathbb{E}(Y^{0,h_1^0,h_2^1} \mid \bar{S}_1, A_0)$. This counterfactual mean is not necessarily equal to $\mathbb{E}(Y^{0,h_1^0,h_2^0} \mid \bar{S}_1, A_0)$ as we argued above, so a direct blipping down would not recover the counterfactual means under the regime in trial 0. To solve this problem, we add a correction term

$$\mathbb{E}\left[\frac{\mathbb{I}\{A_1 = h_1^0(\bar{S}_1)\}}{P\{A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1\}} h_2^0(\bar{S}_2) - h_2^1(\bar{S}_2) \mid S_0, A_0\right],$$

which equals to $\mathbb{E}(Y^{A_0,h_1^0,h_2^0} - Y^{A_0,h_1^0,h_2^1} \mid S_0, A_0)$.

We now give a class of influence functions for the parameter ψ in $\gamma_0(S_0; \psi)$. For simplicity, we treat $\gamma_1(\bar{S}_1)$ and $\gamma_2(\bar{S}_2)$ as known and consider influence functions constructed from the Trial 0 moment condition.

Theorem 4. *For P satisfying Conditions 3 5, 4, and conditions in Theorem 1. For any measurable function $c : \mathcal{S}_0 \rightarrow \mathbb{R}^p$, define*

$$\begin{aligned} \pi_0(S_0) &= \mathbb{P}(A_0 = 1 \mid S_0), \\ \mu_0^0(S_0) &= \mathbb{E}(\mathbb{E}[\mathbb{E}\{Y \mid \bar{S}_2, A_2 = h_2^0(\bar{S}_2)\} \mid \bar{S}_1, A_1 = h_1^0(\bar{S}_1)] \mid S_0, A_0 = 0) \\ q(\bar{S}_1) &= \mathbb{E}\{\Delta\gamma_2(\bar{S}_2) \mid \bar{S}_1, A_1 = h_1^0(\bar{S}_1)\}, \\ \dot{\gamma}_0(S_0) &= \frac{\partial \gamma_0(S_0; \psi)}{\partial \psi} \Big|_{\psi=\psi^*}, \\ V(S_0) &= \mathbb{E}[c(S_0)\pi_0(S_0)\{1 - \pi_0(S_0)\}\dot{\gamma}_0(S_0)^\top]. \end{aligned}$$

Suppose $V(S_0)$ is nonsingular and the expression below is square-integrable, then an influence

function for ψ at ψ^* is

$$\begin{aligned} \varphi_c(O) = & V(S_0)^{-1} c(S_0) \{A_0 - \pi_0(S_0)\} \\ & \times \left[Y - \{A_2 - h_2^1(\bar{S}_2)\} \gamma_2(\bar{S}_2) - \{A_1 - h_1^0(\bar{S}_1)\} \gamma_1(\bar{S}_1) \right. \\ & - A_0 \gamma_0(S_0; \psi^*) + \frac{\mathbb{I}\{A_1 = h_1^0(\bar{S}_1)\}}{P\{A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1\}} \Delta \gamma_2(\bar{S}_2) - \mu_0^0(S_0) \\ & \left. + q(\bar{S}_1) - \frac{\mathbb{I}\{A_1 = h_1^0(\bar{S}_1)\}}{P\{A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1\}} q(\bar{S}_1) \right]. \end{aligned}$$

Consequently, varying $c(s_0)$ over functions satisfying these conditions gives a class of influence functions for ψ .

To give more insights, suppose the causal contrast satisfy the sufficient condition in Theorem 1, i.e.,

$$h_2^0 \{a_0, a_1 = h_1^0\} = h_2^1 \{a_0, a_1 = h_1^0\}.$$

It then follows directly that

$$\frac{\mathbb{I}\{A_1 = h_1^0(\bar{S}_1)\}}{P\{A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1\}} \Delta \gamma_2(\bar{S}_2) = 0, \quad q(\bar{S}_1) = \mathbb{E}\{\Delta \gamma_2(\bar{S}_2) \mid \bar{S}_1, A_1 = h_1^0(\bar{S}_1)\} = 0.$$

The influence function in Theorem 4 then is

$$\begin{aligned} \varphi_c(O) = & V(S_0)^{-1} c(S_0) \{A_0 - \pi_0(S_0)\} \\ & \times \left[Y - \{A_2 - h_2^1(\bar{S}_2)\} \gamma_2(\bar{S}_2) - \{A_1 - h_1^0(\bar{S}_1)\} \gamma_1(\bar{S}_1) - A_0 \gamma_0(S_0; \psi^*) - \mu_0^0(S_0) \right], \end{aligned}$$

which is the influence function for the standard g -estimation. The semiparametric theory leads to the same result as we have derived in Theorem 2.

Proof. By the third equation in Theorem 3. For any indexing function $c(s_0)$ satisfying the condition in Theorem 4, it follows that

$$\mathbb{E} \left[c_0(S_0) \{A_0 - \pi_0(S_0)\} \{Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) - A_0 \gamma_0(S_0; \psi) + R \Delta \gamma_2(\bar{S}_2)\} \right] = 0$$

We next define P_t as a regular parametric submodel of $\mathcal{M}$, where $P_0 = \mathbb{P}$. We denote $\mathcal{S}(X) \equiv \frac{\partial}{\partial t} P_t(X)|_{t=0}$ the score function for the distribution $P_t(X)$ evaluated at $t = 0$. In what follows, the subscript t refers to evaluation at the law P_t . It then follows that

$$\begin{aligned} & \frac{\partial}{\partial t} \mathbb{E}_t \left[c_0(S_0) \{A_0 - \pi_{0t}(S_0)\} \{Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) \right. \\ & \quad \left. - A_0 \gamma_0(S_0; \psi_t) + R_t \Delta \gamma_2(\bar{S}_2)\} \right] \Big|_{t=0} \\ &= 0 \end{aligned}$$

We have

$$\begin{aligned} & \frac{\partial}{\partial t} \mathbb{E}_t \left[c_0(S_0) \{A_0 - \pi_{0t}(S_0)\} \{Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) \right. \\ & \quad \left. - A_0 \gamma_0(S_0; \psi_t) + R_t \Delta \gamma_2(\bar{S}_2)\} \right] \Big|_{t=0} \\ &= \mathbb{E} \left[c_0(S_0) \{A_0 - \pi_0(S_0)\} \{Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) \right. \\ & \quad \left. - A_0 \gamma_0(S_0; \psi) + R \Delta \gamma_2(\bar{S}_2)\} \mathcal{S}(O) \right] \\ & \quad - \mathbb{E} \left[c_0(S_0) \frac{\partial}{\partial t} \pi_{0t}(S_0) \Big|_{t=0} \{Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) \right. \\ & \quad \left. - A_0 \gamma_0(S_0; \psi_t) + R \Delta \gamma_2(\bar{S}_2)\} \right] \\ & \quad + \mathbb{E} \left[c_0(S_0) \{A_0 - \pi_0(S_0)\} \frac{\partial}{\partial t} R_t \Big|_{t=0} \Delta \gamma_2(\bar{S}_2) \right] \\ & \quad - \mathbb{E} \left[c_0(S_0) \{A_0 - \pi_0(S_0)\} A_0 \dot{\gamma}_0(S_0; \psi) \frac{\partial \psi_t}{\partial t} \Big|_{t=0} \right] \\ &= (i) - (ii) + (iii) - (iv) \\ &= 0 \end{aligned}$$

Terms (i) and (iv) are relatively straightforward and are ready there. We will simply terms (ii) and

(iii). It follows that

$$\begin{aligned}
(ii) &= \mathbb{E} \left[c_0(S_0) \mathbb{E} \{ A_0 \mathcal{S}(A_0 \mid S_0) \mid S_0 \} \{ Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) \right. \\
&\quad \left. - A_0 \gamma_0(S_0; \psi_0) + R \Delta \gamma_2(\bar{S}_2) \} \right] \\
&= \mathbb{E} \left[c_0(S_0) A_0 \mathcal{S}(A_0 \mid S_0) \mu_0^0(S_0) \right] \\
&= \mathbb{E} \left[c_0(S_0) A_0 \mathbb{E} \{ \mathcal{S}(O) \mid A_0, S_0 \} \mu_0^0(S_0) \right] - \mathbb{E} \left[c_0(S_0) A_0 \mathbb{E} \{ \mathcal{S}(O) \mid S_0 \} \mu_0^0(S_0) \right] \\
&= \mathbb{E} \left[c_0(S_0) \{ A_0 - \pi_0(S_0) \} \mu_0^0(S_0) \mathcal{S}(O) \right],
\end{aligned}$$

and

$$\begin{aligned}
(iii) &= -\mathbb{E} \left[c_0(S_0) \{ A_0 - \pi_0(S_0) \} \frac{R}{P \{ A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1 \}} \{ 2h_1(\bar{S}_1) - 1 \} \mathbb{E} \{ A_1 \mathcal{S}(A_1 \mid \bar{S}_1) \mid \bar{S}_1 \} \Delta \gamma_2(\bar{S}_2) \right] \\
&= -\mathbb{E} \left[c_0(S_0) \{ A_0 - \pi_0(S_0) \} \frac{2h_1(\bar{S}_1) - 1}{h_1^0(\bar{S}_1) \pi_1(\bar{S}_1) + \{ 1 - h_1^0(\bar{S}_1) \} \{ 1 - \pi_1(\bar{S}_1) \}} A_1 \mathcal{S}(A_1 \mid \bar{S}_1) q(\bar{S}_1) \right] \\
&= -\mathbb{E} \left[c_0(S_0) \{ A_0 - \pi_0(S_0) \} \frac{2h_1(\bar{S}_1) - 1}{h_1^0(\bar{S}_1) \pi_1(\bar{S}_1) + \{ 1 - h_1^0(\bar{S}_1) \} \{ 1 - \pi_1(\bar{S}_1) \}} \{ A_1 - \pi_1(\bar{S}_1) \} q(\bar{S}_1) \mathcal{S}(O) \right] \\
&= \mathbb{E} \left(c_0(S_0) \{ A_0 - \pi_0(S_0) \} \left[q(\bar{S}_1) - \frac{\mathbb{I} \{ A_1 = h_1^0(\bar{S}_1) \}}{P \{ A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1 \}} q(\bar{S}_1) \right] \mathcal{S}(O) \right).
\end{aligned}$$

Summing terms (i) to (iv) it follows that

$$\begin{aligned}
&\mathbb{E} \left(c(S_0) \{ A_0 - \pi_0(S_0) \} \left[Y - \{ A_2 - h_2^1(\bar{S}_2) \} \gamma_2(\bar{S}_2) - \{ A_1 - h_1^0(\bar{S}_1) \} \gamma_1(\bar{S}_1) \right. \right. \\
&\quad \left. \left. - A_0 \gamma_0(S_0; \psi^*) + \frac{\mathbb{I} \{ A_1 = h_1^0(\bar{S}_1) \}}{P \{ A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1 \}} \Delta \gamma_2(\bar{S}_2) - \mu_0^0(S_0) \right. \right. \\
&\quad \left. \left. + q(\bar{S}_1) - \frac{\mathbb{I} \{ A_1 = h_1^0(\bar{S}_1) \}}{P \{ A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1 \}} q(\bar{S}_1) \right] \mathcal{S}(O) \right) - V \frac{\partial \psi_t}{\partial t} \Big|_{t=0} \\
&= 0.
\end{aligned}$$

Thus, we get the influence function for ψ with

$$\begin{aligned} \varphi_c(O) = & V(S_0)^{-1} c(S_0) \{A_0 - \pi_0(S_0)\} \\ & \times \left[Y - \{A_2 - h_2^1(\bar{S}_2)\} \gamma_2(\bar{S}_2) - \{A_1 - h_1^0(\bar{S}_1)\} \gamma_1(\bar{S}_1) \right. \\ & - A_0 \gamma_0(S_0; \psi^*) + \frac{\mathbb{I}\{A_1 = h_1^0(\bar{S}_1)\}}{P\{A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1\}} \Delta \gamma_2(\bar{S}_2) - \mu_0^0(S_0) \\ & \left. + q(\bar{S}_1) - \frac{\mathbb{I}\{A_1 = h_1^0(\bar{S}_1)\}}{P\{A_1 = h_1^0(\bar{S}_1) \mid \bar{S}_1\}} q(\bar{S}_1) \right]. \end{aligned}$$

□

8 DISCUSSION

9 APPENDIX

A DYNAMIC REGIMES

We first restrict our attention to meaningful causal contrasts, which are defined as follows.

Condition 7 (Distinct treatment arms). *For each $m = 0, \dots, K$, the two treatment arms in Trial m differ with positive probability; that is,*

$$P \left[\bigcup_{k=m}^K \{g_k^m(\bar{S}_k) \neq g_k^{m'}(\bar{S}_k)\} \right] > 0$$

for every $P \in \mathcal{P}$.

Condition 7 ensures that the two treatment arms in each Trial m are not identical with probability one. In particular, it excludes trials that compare identical treatment assignments for all participants. For a counterexample, consider the three-time example, where we let $g_2^2(\bar{s}_2) = g_2^{2'}(\bar{s}_2) = 1$ for all $\bar{s}_2 \in \bar{\mathcal{S}}_2$. Both counterfactual outcomes in Trial 2 would be $Y_3^{A_0, A_1, 1}$, and their difference is identically zero. This trial would therefore contain no causal contrast of interest and is not considered.

Now we are ready to give a sufficient condition for the causal contrasts to be variation-independent.

Theorem A.1 (A sufficient condition for variation independence). *Suppose that, for each $P \in \mathcal{P}$, the counterfactual outcomes under all treatment arms under consideration are P -integrable, and that $\mathcal{P}$ places no restrictions on the real-valued conditional mean function $\mu_K(\bar{s}_K, a_K) = \mathbb{E}_P(Y_{K+1} \mid \bar{S}_K = \bar{s}_K, A_K = a_K)$. If $g_m^m(\bar{s}_m) \neq g_m^{m'}(\bar{s}_m)$ for every $m = 0, \dots, K$ and every $\bar{s}_m \in \bar{\mathcal{S}}_m$, then $\tau_0, \dots, \tau_K$ are jointly variation-independent; equivalently, $\mathcal{T} = \prod_{m=0}^K \mathcal{T}_m$.*

The proof is given in XX. Theorem A.1 requires that in Trial m , in any stratum defined by $\bar{S}_m$, the two arms should start with different values. Here we do not restrict the type treatment arms and they can be static or dynamic. The next Theorem gives a sufficient and necessary condition for causal contrasts under static regimes to be variation-independent.

From now on, we will only consider causal contrasts of the form

$$\mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}^m} \mid \bar{S}_m = \bar{s}_m \right),$$

which are blip functions that can be elements of an SNMM. The following theorem provides a sufficient condition for them to form an SNMM under dynamic and static regimes.

Theorem A.2. *A sufficient condition for the variation-independent causal contrasts*

$$\mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}^m} \mid \bar{S}_m = \bar{s}_m \right) = \gamma_m^m(P)(\bar{s}_m),$$

with $0 \leq m \leq K$ to form an SNMM for all $P \in \mathcal{P}$ is

$$h_k^m \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \bar{l}_k \} = h_k^j \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \bar{l}_k \},$$

for $0 \leq m < j \leq k-1 \leq K-1$ and all $\bar{a}_m \in \bar{\mathcal{A}}_m, \bar{l}_k \in \bar{\mathcal{L}}_k$.

We note that the index condition $0 \leq m < j \leq k-1 \leq K-1$ applies only when $K \geq 2$. When $K < 2$, the cross-trial compatibility condition is vacuous. We also have a sufficient and necessary condition for causal contrasts under static treatment arms.

for example, when treatment has no effect on the counterfactual outcome and all relevant causal contrasts are zero. For dynamic regimes, however, an additional support issue arises because the continuation treatment functions depend on the time-varying covariate history. Consider a tuple (m, j, k) with $m < j \leq k-1$, and suppose that

$$h_k^m \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \bar{l}_k^* \} \neq h_k^j \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \bar{l}_k^* \}$$

for some covariate history $\bar{l}_k^*$. Under a particular law $P \in \mathcal{P}$, it is possible that the histories on which h_k^m and h_k^j differ are never reached along the treatment path $(\bar{a}_m, h_{m+1:k-1}^m)$. In that case, the two continuation functions may differ as functions while still inducing the same causal

contrasts under P . Thus, for a fixed P , the relevant compatibility condition is naturally an almost-everywhere condition on histories reachable under the corresponding treatment path, rather than pointwise equality of the treatment functions over their entire domains.

Fuyu notes: I think if we are working with all $P \in \mathcal{P}$, we do not need to worry about the time-varying confounder distributions. However, these technical details have not yet been resolved.

Proof of Theorem A.1. By definition, $\mathcal{T} \subseteq \prod_{m=0}^K \mathcal{T}_m$. It therefore remains to show that $\prod_{m=0}^K \mathcal{T}_m \subseteq \mathcal{T}$. To prove that, given an arbitrary tuple $\eta = (\eta_0, \dots, \eta_K) \in \prod_{m=0}^K \mathcal{T}_m$, we will construct a single law $P_\eta \in \mathcal{P}$ such that $\tau_m(P_\eta) = \eta_m$ for every $m = 0, \dots, K$. Throughout the proof, we will fix a reference law P_s for the longitudinal history $(\bar{L}_K, \bar{A}_K)$, and we will vary only the conditional law of Y_{K+1} given $(\bar{S}_K, A_K)$.

For each $m = 0, \dots, K$, define the switch function

$$J_m(\bar{s}_m, a_m) = \mathbb{I}\{a_m = g_m^m(\bar{s}_m)\}.$$

Because $g_m^m(\bar{s}_m) \neq g_m^{m'}(\bar{s}_m)$, we have

$$J_m\{\bar{s}_m, g_m^m(\bar{s}_m)\} = 1 \quad \text{and} \quad J_m\{\bar{s}_m, g_m^{m'}(\bar{s}_m)\} = 0$$

for every $\bar{s}_m \in \bar{\mathcal{S}}_m$.

We construct measurable functions $d_m : \bar{\mathcal{S}}_m \rightarrow \mathbb{R}$ recursively, beginning with $m = K$ and proceeding backward to $m = 0$. Suppose that $d_{m+1}, \dots, d_K$ have already been constructed, and define

$$R_m = \sum_{r=m+1}^K d_r(\bar{S}_r) J_r(\bar{S}_r, A_r),$$

where the empty sum is defined to be zero. In particular, $R_K = 0$. We next use the g-formula to define the contribution of R_m to Trial m contrast. Fix $\bar{s}_m = (\bar{l}_m, \bar{a}_{m-1})$. Under the first arm of

Trial m , we set the subsequent treatment and history paths recursively by

$$a_m^m = g_m^m(\bar{s}_m), \bar{s}_q^m = (\bar{l}_q, \bar{a}_{m-1}, a_m^m, \dots, a_{q-1}^m), a_q^m = g_q^m(\bar{s}_q^m),$$

for $q = m + 1, \dots, K$. Similarly, under the second arm of Trial m , we let

$$a_m^{m'} = g_m^{m'}(\bar{s}_m), \bar{s}_q^{m'} = (\bar{l}_q, \bar{a}_{m-1}, a_m^{m'}, \dots, a_{q-1}^{m'}), a_q^{m'} = g_q^{m'}(\bar{s}_q^{m'}),$$

for $q = m + 1, \dots, K$. We let

$$R_m^m = \sum_{r=m+1}^K d_r(\bar{s}_r^m) J_r(\bar{s}_r^m, a_r^m), \quad R_m^{m'} = \sum_{r=m+1}^K d_r(\bar{s}_r^{m'}) J_r(\bar{s}_r^{m'}, a_r^{m'})$$

denote the values of R_m along the covariate and treatment paths induced by the first and second arms, respectively. Let

$$\begin{aligned} \kappa_m(\bar{s}_m) = & \int \cdots \int R_m^m \prod_{q=m+1}^K dP_s(l_q \mid \bar{S}_{q-1} = \bar{s}_{q-1}^m, A_{q-1} = a_{q-1}^m) \\ & - \int \cdots \int R_m^{m'} \prod_{q=m+1}^K dP_s(l_q \mid \bar{S}_{q-1} = \bar{s}_{q-1}^{m'}, A_{q-1} = a_{q-1}^{m'}), \end{aligned}$$

where both integrals are taken over $l_{m+1}, \dots, l_K$. Note that $\kappa_m(\bar{s}_m)$ is the Trial m contrast with later treatment trajectory determined by the later trial arms. One can show that under P_s , $\kappa_m(\bar{s}_m)$ is determined by future later functions $d_{m+1}, \dots, d_K$. Then, we set

$$d_m(\bar{s}_m) = \eta_m(\bar{s}_m) - \kappa_m(\bar{s}_m).$$

This recursion is well defined because κ_m depends only on the previously constructed functions $d_{m+1}, \dots, d_K$. At time K , there are no later terms, so $R_K = 0$ and $\kappa_K = 0$. Therefore, $d_K = \eta_K$. Once d_K has been chosen, we can calculate κ_{K-1} and define $d_{K-1} = \eta_{K-1} - \kappa_{K-1}$. Continuing backward in this way defines d_m for every $m = 0, \dots, K$.

Having constructed $d_0, \dots, d_K$, define the expectation of Y_{K+1} conditional on $\bar{S}_K, A_K$ as,

$$m_\eta(\bar{S}_K, A_K) = \sum_{r=0}^K d_r(\bar{S}_r) J_r(\bar{S}_r, A_r).$$

The function m_η is measurable with respect to $(\bar{S}_K, A_K)$. Under the assumption that the conditional mean of Y_{K+1} is unrestricted over the class of integrable real-valued functions, define P_η by retaining the longitudinal distribution under P_s and setting $\mathbb{E}_{P_\eta}(Y_{K+1} \mid \bar{S}_K, A_K) = m_\eta(\bar{S}_K, A_K)$. For example, one can choose

$$Y_{K+1} \mid \bar{S}_K, A_K \sim N \{m_\eta(\bar{S}_K, A_K), 1\},$$

and construct $P_\eta = P_s \times P_{Y_{K+1} \mid \bar{S}_K, A_K}$. We now verify that $\tau_m(P_\eta) = \eta_m$ for every $m = 0, \dots, K$. Fix $\bar{s}_m = (\bar{l}_m, \bar{a}_{m-1})$. By consistency, sequential exchangeability, positivity, and the g-formula,

$$\begin{aligned} \tau_m(P_\eta)(\bar{s}_m) &= \int \cdots \int m_\eta(\bar{s}_K^m, a_K^m) \prod_{q=m+1}^K dP_s(l_q \mid \bar{S}_{q-1} = \bar{s}_{q-1}^m, A_{q-1} = a_{q-1}^m) \\ &\quad - \int \cdots \int m_\eta(\bar{s}_K^{m'}, a_K^{m'}) \prod_{q=m+1}^K dP_s(l_q \mid \bar{S}_{q-1} = \bar{s}_{q-1}^{m'}, A_{q-1} = a_{q-1}^{m'}) . \end{aligned}$$

Along the treatment and covariate path induced by the first arm of Trial m , the terminal conditional mean satisfies

$$\begin{aligned} m_\eta(\bar{s}_K^m, a_K^m) &= \sum_{r=0}^{m-1} d_r(\bar{s}_r) J_r(\bar{s}_r, a_r) + d_m(\bar{s}_m) J_m \{ \bar{s}_m, g_m^m(\bar{s}_m) \} + \sum_{r=m+1}^K d_r(\bar{s}_r^m) J_r(\bar{s}_r^m, a_r^m). \\ &= \sum_{r=0}^{m-1} d_r(\bar{s}_r) J_r(\bar{s}_r, a_r) + d_m(\bar{s}_m) + R_m^m, \end{aligned}$$

The second equality follows because $J_m \{ \bar{s}_m, g_m^m(\bar{s}_m) \} = 1$. Similarly, along the path induced by

the second arm,

$$\begin{aligned} m_\eta(\bar{s}_K^{m,'}, a_K^{m,'}) &= \sum_{r=0}^{m-1} d_r(\bar{s}_r) J_r(\bar{s}_r, a_r) + d_m(\bar{s}_m) J_m\{\bar{s}_m, g_m^{m,'}(\bar{s}_m)\} + \sum_{r=m+1}^K d_r(\bar{s}_r^{m,'}) J_r(\bar{s}_r^{m,'}, a_r^{m,'}) \\ &= \sum_{r=0}^{m-1} d_r(\bar{s}_r) J_r(\bar{s}_r, a_r) + R_m^{m,'}, \end{aligned}$$

The second equality follows because $g_m^m(\bar{s}_m) \neq g_m^{m,'}(\bar{s}_m)$ and hence $J_m\{\bar{s}_m, g_m^{m,'}(\bar{s}_m)\} = 0$.

Substituting these expressions into the g-formula gives

$$\begin{aligned} \tau_m(P_\eta)(\bar{s}_m) &= \int \cdots \int \left\{ \sum_{r=0}^{m-1} d_r(\bar{s}_r) J_r(\bar{s}_r, a_r) + d_m(\bar{s}_m) + R_m^m \right\} \\ &\quad \times \prod_{q=m+1}^K dP_s(l_q \mid \bar{S}_{q-1} = \bar{s}_{q-1}^m, A_{q-1} = a_{q-1}^m) \\ &\quad - \int \cdots \int \left\{ \sum_{r=0}^{m-1} d_r(\bar{s}_r) J_r(\bar{s}_r, a_r) + R_m^{m,'} \right\} \\ &\quad \times \prod_{q=m+1}^K dP_s(l_q \mid \bar{S}_{q-1} = \bar{s}_{q-1}^{m'}, A_{q-1} = a_{q-1}^{m'}). \end{aligned}$$

For $r < m$, the quantities $d_r(\bar{s}_r) J_r(\bar{s}_r, a_r)$ are fixed by the conditioned history $\bar{s}_m$. Therefore, their integrals under either treatment arm equal the same quantities and cancel. Moreover, $d_m(\bar{s}_m)$ is fixed given $\bar{s}_m$, so its integral under the first arm equals $d_m(\bar{s}_m)$. It follows that

$$\begin{aligned} \tau_m(P_\eta)(\bar{s}_m) &= d_m(\bar{s}_m) \\ &\quad + \int \cdots \int R_m^m \prod_{q=m+1}^K dP_s(l_q \mid \bar{S}_{q-1} = \bar{s}_{q-1}^m, A_{q-1} = a_{q-1}^m) \\ &\quad - \int \cdots \int R_m^{m,'} \prod_{q=m+1}^K dP_s(l_q \mid \bar{S}_{q-1} = \bar{s}_{q-1}^{m'}, A_{q-1} = a_{q-1}^{m'}) \\ &= d_m(\bar{s}_m) + \kappa_m(\bar{s}_m) \\ &= \eta_m(\bar{s}_m), \end{aligned}$$

where the second equality follows from the definition of κ_m , and the final equality follows from $d_m(\bar{s}_m) = \eta_m(\bar{s}_m) - \kappa_m(\bar{s}_m)$. Because this equality holds for every $\bar{s}_m \in \bar{\mathcal{S}}_m$, we have $\tau_m(P_\eta) = \eta_m$. Therefore, the causal contrast maps $\tau_0, \dots, \tau_K$ are jointly variation-independent. $\square$

Proof of Theorem 1. The sufficiency follows directly from Theorem A.1. We prove the necessity by contradiction. Suppose that $\tau_0, \dots, \tau_K$ are jointly variation-independent, but that the stated condition does not hold. Then there exists at least one time m and one relevant treatment history $\bar{a}_{m-1}$ such that $a_m^m(\bar{a}_{m-1}) = a_m^{m'}(\bar{a}_{m-1})$. Let

$$m^\dagger = \max \{m : \text{there exists a } \bar{a}_{m-1} \text{ such that } a_m^m(\bar{a}_{m-1}) = a_m^{m'}(\bar{a}_{m-1})\}.$$

Fix one such history $\bar{a}_{m^\dagger-1}$ and write

$$a_{m^\dagger}^{m^\dagger}(\bar{a}_{m^\dagger-1}) = a_{m^\dagger}^{m^\dagger'}(\bar{a}_{m^\dagger-1}) = a_{m^\dagger}.$$

By the maximality of $m^\dagger$, for every $r = m^\dagger + 1, \dots, K$ and every relevant treatment history $\bar{a}_{r-1}$, $a_r^r(\bar{a}_{r-1}) \neq a_r^{r'}(\bar{a}_{r-1})$. Moreover, by the assumed stratum-specific distinctness of the complete Trial $m^\dagger$ arms,

$$\underline{a}_{m^\dagger}^{m^\dagger}(\bar{a}_{m^\dagger-1}) \neq \underline{a}_{m^\dagger}^{m^\dagger'}(\bar{a}_{m^\dagger-1}).$$

Since the two arms assign the same treatment at time $m^\dagger$, they must therefore differ at some later treatment time. For $r = m^\dagger + 1, \dots, K$, let $0 \in \mathcal{T}_r$ denote the identically zero contrast function. Consider any law $P \in \mathcal{P}$ satisfying $\tau_r(P) = 0$ for all $r = m^\dagger + 1, \dots, K$.

By Lemma 1, the zero contrasts in all later trials, together with $a_r^r(\bar{a}_{r-1}) \neq a_r^{r'}(\bar{a}_{r-1})$ for $r > m^\dagger$, imply that all static treatment arms beginning at time $m^\dagger + 1$ have the same conditional counterfactual mean. That is, for any two static continuation regimes $\underline{b}_{m^\dagger+1}$ and $\underline{c}_{m^\dagger+1}$,

$$\mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m^\dagger}, \underline{b}_{m^\dagger+1}} \mid \bar{S}_{m^\dagger+1} = \bar{s}_{m^\dagger+1} \right) = \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m^\dagger}, \underline{c}_{m^\dagger+1}} \mid \bar{S}_{m^\dagger+1} = \bar{s}_{m^\dagger+1} \right)$$

for every relevant $\bar{s}_{m^\dagger+1}$.

Now fix any $\bar{s}_{m^\dagger}$ whose treatment component is $\bar{a}_{m^\dagger-1}$. Since both arms of Trial $m^\dagger$ assign $A_{m^\dagger} = a_{m^\dagger}$, the g-formula gives

$$\begin{aligned} \{\tau_{m^\dagger}(P)\}(\bar{s}_{m^\dagger}) &= \int \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m^\dagger}, a_{m^\dagger+1}^{m^\dagger}} \mid \bar{S}_{m^\dagger+1} = (\bar{s}_{m^\dagger}, a_{m^\dagger}, l_{m^\dagger+1}) \right) \\ &\quad \times dP(l_{m^\dagger+1} \mid \bar{S}_{m^\dagger} = \bar{s}_{m^\dagger}, A_{m^\dagger} = a_{m^\dagger}) \\ &\quad - \int \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{m^\dagger}, a_{m^\dagger+1}^{m^\dagger, \prime}} \mid \bar{S}_{m^\dagger+1} = (\bar{s}_{m^\dagger}, a_{m^\dagger}, l_{m^\dagger+1}) \right) \\ &\quad \times dP(l_{m^\dagger+1} \mid \bar{S}_{m^\dagger} = \bar{s}_{m^\dagger}, A_{m^\dagger} = a_{m^\dagger}). \end{aligned}$$

By Lemma 1, the two conditional means inside the integrals are equal for every relevant value of $l_{m^\dagger+1}$. Therefore,

$$\{\tau_{m^\dagger}(P)\}(\bar{s}_{m^\dagger}) = 0.$$

Hence, for every $P \in \mathcal{P}$,

$$\tau_{m^\dagger+1}(P) = 0_{m^\dagger+1}, \dots, \tau_K(P) = 0_K \implies \{\tau_{m^\dagger}(P)\}(\bar{s}_{m^\dagger}) = 0$$

for every $\bar{s}_{m^\dagger}$ whose treatment component is $\bar{a}_{m^\dagger-1}$. We next show that $\mathcal{T}_{m^\dagger}$ contains a contrast function that is nonzero on this stratum. Because the two complete treatment arms of Trial $m^\dagger$ are distinct, define

$$h^\dagger(\bar{S}_K, A_K) = \mathbb{I} \{ \bar{A}_{m^\dagger-1} = \bar{a}_{m^\dagger-1} \} \mathbb{I} \{ A_{m^\dagger:K} = \underline{a}_{m^\dagger}^{m^\dagger}(\bar{a}_{m^\dagger-1}) \}.$$

By the assumption that the conditional mean of Y_{K+1} is unrestricted, there exists a law $P^\dagger \in \mathcal{P}$

such that

$$\mathbb{E}_{P^\dagger}(Y_{K+1} \mid \bar{S}_K, A_K) = h^\dagger(\bar{S}_K, A_K).$$

For every relevant $\bar{s}_{m^\dagger}$ whose treatment component is $\bar{a}_{m^\dagger-1}$, the first arm of Trial $m^\dagger$ follows $\underline{a}_{m^\dagger}^{m^\dagger}(\bar{a}_{m^\dagger-1})$, so that $h^\dagger = 1$. The second arm follows the distinct treatment sequence $\underline{a}_{m^\dagger}^{m^\dagger, '}'(\bar{a}_{m^\dagger-1})$, so that $h^\dagger = 0$. Therefore, $\{\tau_{m^\dagger}(P^\dagger)\}(\bar{s}_{m^\dagger}) = 1$. Let $\eta_{m^\dagger}^\dagger = \tau_{m^\dagger}(P^\dagger)$. Then $\eta_{m^\dagger}^\dagger \in \mathcal{T}_{m^\dagger}$ and

$$\eta_{m^\dagger}^\dagger(\bar{s}_{m^\dagger}) = 1$$

on the stratum with treatment history $\bar{a}_{m^\dagger-1}$.

For each $r = m^\dagger + 1, \dots, K$, the identically zero function belongs to $\mathcal{T}_r$, since it can be obtained, for example, under a law with a constant conditional mean for Y_{K+1} . Now suppose that $\tau_0, \dots, \tau_K$ are jointly variation-independent. Choose arbitrary $\eta_r \in \mathcal{T}_r$ for $r = 0, \dots, m^\dagger - 1$. By joint variation independence, there must exist a single law $\tilde{P} \in \mathcal{P}$ such that

$$\begin{aligned} \tau_r(\tilde{P}) &= \eta_r, & r &= 0, \dots, m^\dagger - 1, \\ \tau_{m^\dagger}(\tilde{P}) &= \eta_{m^\dagger}^\dagger, \\ \tau_r(\tilde{P}) &= 0, & r &= m^\dagger + 1, \dots, K. \end{aligned}$$

However, we have already shown that, under any law $P \in \mathcal{P}$,

$$\tau_{m^\dagger+1}(P) = \dots = \tau_K(P) = 0 \implies \{\tau_{m^\dagger}(P)\}(\bar{s}_{m^\dagger}) = 0$$

for every $\bar{s}_{m^\dagger}$ whose treatment component is $\bar{a}_{m^\dagger-1}$. Applying this result to $\tilde{P}$ gives

$$\{\tau_{m^\dagger}(\tilde{P})\}(\bar{s}_{m^\dagger}) = 0.$$

But $\tau_{m^\dagger}(\tilde{P}) = \eta_{m^\dagger}^\dagger$, and by construction

$$\eta_{m^\dagger}^\dagger(\bar{s}_{m^\dagger}) = 1.$$

This is a contradiction.

Therefore, $a_m^m(\bar{a}_{m-1}) \neq a_m^{m'}(\bar{a}_{m-1})$ for every $m = 0, \dots, K$ and every relevant treatment history $\bar{a}_{m-1}$. This proves the necessity. $\square$

Lemma 1 (Equality of static treatment arm means). *Fix $P \in \mathcal{P}$ and $q \in \{0, \dots, K\}$, and suppose that $A_r \in \{0, 1\}$ for $r = q, \dots, K$. For any static treatment arm $\underline{b}_q = (b_q, \dots, b_K)$, define*

$$\mu_q^{\underline{b}_q}(\bar{s}_q) = \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{q-1}, \underline{b}_q} \mid \bar{S}_q = \bar{s}_q \right).$$

Suppose that, for every $r = q, \dots, K$, the causal contrast between the two pre-specified treatment arms of Trial r is identically zero; that is,

$$\{\tau_r(P)\}(\bar{s}_r) = \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{r-1}, \underline{a}_r} \mid \bar{S}_r = \bar{s}_r \right) - \mathbb{E}_P \left(Y_{K+1}^{\bar{A}_{r-1}, \underline{a}_r'} \mid \bar{S}_r = \bar{s}_r \right) = 0$$

for every relevant $\bar{s}_r \in \bar{\mathcal{S}}_r$. Suppose further that

$$a_r^r(\bar{a}_{r-1}) \neq a_r^{r'}(\bar{a}_{r-1})$$

for every $r = q, \dots, K$ and every relevant treatment history $\bar{a}_{r-1}$.

Then, for any two static treatment arms $\underline{b}_q = (b_q, \dots, b_K)$ and $\underline{c}_q = (c_q, \dots, c_K)$,

$$\mu_q^{\underline{b}_q}(\bar{s}_q) = \mu_q^{\underline{c}_q}(\bar{s}_q)$$

for every relevant $\bar{s}_q \in \bar{\mathcal{S}}_q$.

Proof. We prove the statement for $q = K, K-1, \dots, 0$ by backward induction.

For $q = K$, a static treatment arm consists of a single treatment value $b_K \in \{0, 1\}$. Because

$$a_K^K(\bar{a}_{K-1}) \neq a_K^{K'}(\bar{a}_{K-1})$$

and A_K is binary, we have

$$\left\{ a_K^K(\bar{a}_{K-1}), a_K^{K'}(\bar{a}_{K-1}) \right\} = \{0, 1\}.$$

Therefore, $\{\tau_K(P)\}(\bar{s}_K) = 0$ implies

$$\mu_K^0(\bar{s}_K) = \mu_K^1(\bar{s}_K)$$

for every relevant $\bar{s}_K$. Hence, the result holds for $q = K$.

Now suppose that the result holds at time $q + 1$. Consider any two static treatment arms

$$\underline{b}_q = (b_q, \underline{b}_{q+1}) \quad \text{and} \quad \underline{c}_q = (c_q, \underline{c}_{q+1}),$$

and fix a relevant $\bar{s}_q = (\bar{l}_q, \bar{a}_{q-1})$.

First, consider the case $b_q = c_q$. By the g-formula,

$$\mu_q^{(b_q, \underline{b}_{q+1})}(\bar{s}_q) = \int \mu_{q+1}^{\underline{b}_{q+1}}(\bar{s}_q, b_q, l_{q+1}) dP(l_{q+1} \mid \bar{S}_q = \bar{s}_q, A_q = b_q),$$

and

$$\begin{aligned} \mu_q^{(c_q, \underline{c}_{q+1})}(\bar{s}_q) &= \int \mu_{q+1}^{\underline{c}_{q+1}}(\bar{s}_q, c_q, l_{q+1}) dP(l_{q+1} \mid \bar{S}_q = \bar{s}_q, A_q = c_q) \\ &= \int \mu_{q+1}^{\underline{c}_{q+1}}(\bar{s}_q, b_q, l_{q+1}) dP(l_{q+1} \mid \bar{S}_q = \bar{s}_q, A_q = b_q), \end{aligned}$$

where the second equality follows from $b_q = c_q$.

By the induction hypothesis,

$$\mu_{q+1}^{b_{q+1}}(\bar{s}_{q+1}) = \mu_{q+1}^{c_{q+1}}(\bar{s}_{q+1})$$

for every relevant $\bar{s}_{q+1} \in \bar{\mathcal{S}}_{q+1}$. Therefore, the integrands in the preceding two expressions are equal for every relevant l_{q+1} , and hence

$$\mu_q^{b_q}(\bar{s}_q) = \mu_q^{c_q}(\bar{s}_q).$$

Next, consider the case $b_q \neq c_q$. Since A_q is binary,

$$\{b_q, c_q\} = \{0, 1\}.$$

Similarly, because $a_q^q(\bar{a}_{q-1}) \neq a_q^{q'}(\bar{a}_{q-1})$,

$$\{a_q^q(\bar{a}_{q-1}), a_q^{q'}(\bar{a}_{q-1})\} = \{0, 1\}.$$

It follows that the ordered pair (b_q, c_q) is either

$$(a_q^q(\bar{a}_{q-1}), a_q^{q'}(\bar{a}_{q-1}))$$

or

$$(a_q^{q'}(\bar{a}_{q-1}), a_q^q(\bar{a}_{q-1})).$$

Let $\underline{a}_{q+1}^{q,b}$ denote the continuation from time $q+1$ onward of the Trial q arm that assigns b_q at time q . Similarly, let $\underline{a}_{q+1}^{q,c}$ denote the continuation of the Trial q arm that assigns c_q at time q .

By the g-formula,

$$\mu_q^{(b_q, b_{q+1})}(\bar{s}_q) = \int \mu_{q+1}^{b_{q+1}}(\bar{s}_q, b_q, l_{q+1}) dP(l_{q+1} \mid \bar{S}_q = \bar{s}_q, A_q = b_q).$$

By the induction hypothesis, once $A_q = b_q$ is fixed, any two static treatment arms from time $q + 1$ onward have the same conditional counterfactual mean. Therefore,

$$\mu_{q+1}^{b_{q+1}}(\bar{s}_q, b_q, l_{q+1}) = \mu_{q+1}^{a_{q+1}^{q,b}}(\bar{s}_q, b_q, l_{q+1}),$$

and consequently

$$\mu_q^{(b_q, b_{q+1})}(\bar{s}_q) = \mu_q^{(b_q, a_{q+1}^{q,b})}(\bar{s}_q).$$

Similarly,

$$\mu_q^{(c_q, c_{q+1})}(\bar{s}_q) = \mu_q^{(c_q, a_{q+1}^{q,c})}(\bar{s}_q).$$

The two expressions on the right-hand sides are precisely the conditional counterfactual means under the two pre-specified arms of Trial q , possibly in the reverse order. Since $\{\tau_q(P)\}(\bar{s}_q) = 0$, these two conditional means are equal. Therefore,

$$\mu_q^{b_q}(\bar{s}_q) = \mu_q^{c_q}(\bar{s}_q).$$

The result follows by backward induction. □

Proof of Proposition 1. By Assumptions ?? and ??

$$\begin{aligned}
& \mathbb{E}_P \left(Y^{\bar{A}_m, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, \underline{h}_m} \mid \bar{S}_m = \bar{s}_m, A_m = 1 \right) \\
&= \mathbb{E} \left(Y^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, h_m, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m, A_m = 1 \right) \\
&= \mathbb{E} \left(Y^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, h_m, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m \right) \\
&= (1 - h_m) \gamma_m(\bar{s}_m).
\end{aligned}$$

On the stratum $\bar{s}_m$ such that $h_m(\bar{s}_m) = 1$, both sides of the above equation equal to 0. On the stratum $\bar{s}_m$ such that $h_m(\bar{s}_m) = 0$, it then follows that $\mathbb{E} \left(Y^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m \right) = \gamma_m(\bar{s}_m)$. Similarly, we have

$$\begin{aligned}
& \mathbb{E} \left(Y^{\bar{A}_m, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, \underline{h}_m} \mid \bar{S}_m = \bar{s}_m, A_m = 0 \right) \\
&= \mathbb{E} \left(Y^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, h_m, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m, A_m = 0 \right) \\
&= \mathbb{E} \left(Y^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, h_m, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m \right) \\
&= -h_m \gamma_m(\bar{s}_m).
\end{aligned}$$

On the stratum $\bar{s}_m$ such that $h_m(\bar{s}_m) = 0$, both sides of the above equation equal to 0. On the stratum $\bar{s}_m$ such that $h_m(\bar{s}_m) = 1$, it then follows that $\mathbb{E} \left(Y^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m \right) = -\gamma_m(\bar{s}_m)$. Rearranging the order of terms, it leads to $\mathbb{E} \left(Y^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}} - Y^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m \right) = \gamma_m(\bar{s}_m)$. Because $h_m(\bar{s}_m)$ can only take values on $\{0, 1\}$, we have the results on all h_m and thus all $\bar{s}_m$. $\square$

Proof of Proposition 2. Because A_m is binary and $g_m^m(\bar{s}_m) \neq g_m^{m'}(\bar{s}_m)$, we have

$$g_m^{m'}(\bar{s}_m) = 1 - g_m^m(\bar{s}_m)$$

for every $\bar{s}_m \in \bar{S}_m$.

For $j = m + 1, \dots, K$, the history $\bar{s}_j$ contains both $\bar{s}_m$ and a_m . Moreover, exactly one of the

indicators

$$\mathbb{I}\{a_m = g_m^m(\bar{s}_m)\} \quad \text{and} \quad \mathbb{I}\{a_m = 1 - g_m^m(\bar{s}_m)\}$$

equals one. Therefore, h_j^m is a well-defined treatment regime taking values in $\{0, 1\}$.

Under the intervention that assigns $A_m = g_m^m(\bar{S}_m)$, we have

$$\mathbb{I}\{A_m = g_m^m(\bar{S}_m)\} = 1 \quad \text{and} \quad \mathbb{I}\{A_m = 1 - g_m^m(\bar{S}_m)\} = 0.$$

It follows that

$$h_j^m(\bar{S}_j) = g_j^m(\bar{S}_j), \quad j = m + 1, \dots, K.$$

Hence, the intervention $(g_m^m, \underline{h}_{m+1}^m)$ induces the same treatment trajectory as the original first arm $(g_m^m, \underline{g}_{m+1}^m)$. Therefore,

$$Y_{K+1}^{\bar{A}_{m-1}, g_m^m, \underline{h}_{m+1}^m} = Y_{K+1}^{\bar{A}_{m-1}, g_m^m, \underline{g}_{m+1}^m}.$$

Similarly, under the intervention that assigns $A_m = 1 - g_m^m(\bar{S}_m) = g_m^{m,'}(\bar{S}_m)$, we have

$$\mathbb{I}\{A_m = g_m^m(\bar{S}_m)\} = 0 \quad \text{and} \quad \mathbb{I}\{A_m = 1 - g_m^m(\bar{S}_m)\} = 1.$$

Thus,

$$h_j^m(\bar{S}_j) = g_j^{m,'}(\bar{S}_j), \quad j = m + 1, \dots, K,$$

and consequently

$$Y_{K+1}^{\bar{A}_{m-1}, 1 - g_m^m, \underline{h}_{m+1}^m} = Y_{K+1}^{\bar{A}_{m-1}, g_m^{m,'}, \underline{g}_{m+1}^{m,'}}.$$

Subtracting the preceding two equalities and taking conditional expectations given $\bar{S}_m = \bar{s}_m$ gives

$$\begin{aligned} & \mathbb{E} \left(Y_{K+1}^{\bar{A}_{m-1}, g_m^m, \underline{g}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, g_m^{m'}, \underline{g}_{m+1}^{m'}} \mid \bar{S}_m = \bar{s}_m \right) \\ &= \mathbb{E} \left(Y_{K+1}^{\bar{A}_{m-1}, g_m^m, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, 1-g_m^m, \underline{h}_{m+1}^m} \mid \bar{S}_m = \bar{s}_m \right). \end{aligned}$$

When $m = K$, the continuation regimes are empty and the same conclusion follows directly from $g_K^{K'} = 1 - g_K^K$. $\square$

Proof of Theorem A.2. Fix $j \in \{1, \dots, K-1\}$ and an earlier trial $m < j$. We first show that, after following the Trial m regime through time j , the continuation regime $\underline{h}_{j+1}^m$ can be replaced by $\underline{h}_{j+1}^j$ without changing the resulting counterfactual outcome.

Fix $a_m \in \{0, 1\}$ and a treatment history generated by $h_{m+1}^m, \dots, h_j^m$. We show that

$$Y_{K+1}^{\bar{a}_{m-1}, a_m, h_{m+1:j}^m, \underline{h}_{j+1}^m} = Y_{K+1}^{\bar{a}_{m-1}, a_m, h_{m+1:j}^m, \underline{h}_{j+1}^j}.$$

We prove that the two continuation regimes assign the same treatment at every time $k = j + 1, \dots, K$. At time $k = j + 1$, the assumed condition gives

$$h_{j+1}^m \{ \bar{a}_m, a_{m+1:j} = h_{m+1:j}^m, \bar{l}_{j+1} \} = h_{j+1}^j \{ \bar{a}_m, a_{m+1:j} = h_{m+1:j}^m, \bar{l}_{j+1} \}.$$

Now suppose that the two continuation regimes have assigned the same treatments from time $j + 1$ through time $k - 1$. They therefore generate the same treatment history up to time k . By the assumed condition,

$$h_k^m \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \bar{l}_k \} = h_k^j \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \bar{l}_k \}.$$

Hence, the two regimes also assign the same treatment at time k . Proceeding recursively for $k = j + 1, \dots, K$ shows that $\underline{h}_{j+1}^m$ and $\underline{h}_{j+1}^j$ generate the same treatment trajectory after time j .

Therefore,

$$Y_{K+1}^{\bar{a}_{m-1}, a_m, h_{m+1:j}^m, \underline{h}_{j+1}^m} = Y_{K+1}^{\bar{a}_{m-1}, a_m, h_{m+1:j}^m, \underline{h}_{j+1}^j}.$$

In particular, this equality holds for both $a_m = 1$ and $a_m = 0$. Therefore, in the Trial m causal contrast, the continuation regime $\underline{h}_{j+1}^m$ can be replaced by $\underline{h}_{j+1}^j$. Because this argument holds for every $m < j$, all trials beginning before j may use the same continuation regime from time $j + 1$ onward. Applying this result successively for $j = 1, \dots, K - 1$ yields a single common regime $\underline{h}$ such that, for every $m = 0, \dots, K$,

$$\mathbb{E} \left(Y_{K+1}^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}^m} \mid \bar{S}_m = \bar{s}_m \right) = \mathbb{E} \left(Y_{K+1}^{\bar{A}_{m-1}, 1, \underline{h}_{m+1}} - Y_{K+1}^{\bar{A}_{m-1}, 0, \underline{h}_{m+1}} \mid \bar{S}_m = \bar{s}_m \right).$$

Hence, all of the causal contrasts can be represented using the same treatment regime $\underline{h}$ and therefore form the blip functions of an h -specific SNMM. $\square$

Proof of Theorem 2. The sufficiency follows directly from Theorem A.2. We prove the necessity by contradiction. Suppose that there exist a tuple (m, j, k) , $m < j \leq k - 1$ such that

$$h_k^m \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \} \neq h_k^j \{ \bar{a}_m, a_{m+1:k-1} = h_{m+1:k-1}^m, \}.$$

Then, the counterfactual outcomes in the following blip function

$$\mathbb{E}_P(Y_{K+1}^{\bar{a}_{m-1}, 1, \underline{h}_{m+1}^m} - Y_{K+1}^{\bar{a}_{m-1}, 0, \underline{h}_{m+1}^m} \mid \bar{S}_m)$$

follow different treatment values from the counterfactual outcomes in the following blip function

$$\mathbb{E}_P(Y_{K+1}^{\bar{a}_{j-1}, 1, \underline{h}_{j+1}^j} - Y_{K+1}^{\bar{a}_{j-1}, 0, \underline{h}_{j+1}^j} \mid \bar{S}_j),$$

at time t_k

Suppose the two blip functions are part of an SNMM, then it follows that with treatment history

$\bar{A}_{k-1} = (\bar{a}_{m-1}, 1, a_{m+1:k-1})$, the two regimes h_k^j and h_k^m assigns the same values, contradicting with the condition. Thus, they are not part of an SNMM.

□

Proof of Theorem 3. The proof for the first two rows are standard results from the g -estimation literature and easy to show here. It suffices to show that

$$\mathbb{E} \{ Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) + R \Delta \gamma_2(\bar{S}_2) \mid \bar{S}_1 \} = \mathbb{E} \left(Y^{A_0, h_1^0, h_2^0} \mid \bar{S}_1 \right).$$

By Proposition 1

$$\mathbb{E}(Y_3^{A_0, A_1, 1} - Y_3^{A_0, A_1, 0} \mid \bar{S}_2) = \gamma_2(\bar{S}_2)$$

implies that

$$\mathbb{E}(Y_3^{A_0, A_1, A_2} - Y_3^{A_0, A_1, h_2^1} \mid \bar{S}_2, A_2) = \{A_2 - h_2^1(\bar{S}_2)\} \gamma_2(\bar{S}_2),$$

which is an usual SNMM with blip function to be h_2^1 . Using the results in the SNMM,

$$\mathbb{E} \{ Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) \mid \bar{S}_2 \} = \mathbb{E}(Y^{A_0, A_1, h_2^1} \mid \bar{S}_2).$$

Then using the same trick on the causal contrast in Trial 1,

$$\mathbb{E}(Y_3^{A_0, 1, h_2^1} - Y_3^{A_0, 0, h_2^1} \mid \bar{S}_1) = \gamma_1(\bar{S}_1)$$

implies that

$$\mathbb{E}(Y_3^{A_0, A_1, h_2^1} - Y_3^{A_0, h_1^0, h_2^1} \mid \bar{S}_1) = \{A_1 - h_1^0(\bar{S}_1)\} \gamma_1(\bar{S}_1).$$

Thus,

$$\begin{aligned}
& \mathbb{E} \left\{ Y - (A_2 - h_2^1) \gamma_2(\bar{S}_2) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) \mid \bar{S}_1, A_1 \right\} \\
&= \mathbb{E}(Y^{A_0, A_1, h_2^1} \mid \bar{S}_1, A_1) - (A_1 - h_1^0) \gamma_1(\bar{S}_1) \\
&= \mathbb{E} \left(Y^{A_0, h_1^0, h_2^1} \mid \bar{S}_1, A_1 \right) \\
&= \mathbb{E} \left(Y^{A_0, h_1^0, h_2^1} \mid \bar{S}_1 \right).
\end{aligned}$$

To show the result holds, the task then is to show that

$$\mathbb{E} \left\{ R \Delta \gamma_2(\bar{S}_2) \mid \bar{S}_1 \right\} = \mathbb{E} \left(Y^{A_0, h_1^0, h_2^0} \mid \bar{S}_1 \right) - \mathbb{E} \left(Y^{A_0, h_1^0, h_2^1} \mid \bar{S}_1 \right).$$

The key observation is

$$\left\{ h_2^1(\bar{S}_2) - h_2^0(\bar{S}_2) \right\} \gamma_2(\bar{S}_2) = \mathbb{E} \left(Y_3^{A_0, A_1, h_2^1} - Y_3^{A_0, A_1, h_2^0} \mid \bar{S}_2 \right),$$

which holds because If $h_2^1(\bar{S}_2) = h_2^0(\bar{S}_2)$, the left-hand-side and the right-hand-side are both zero.

If $h_2^1(\bar{S}_2) = 1$, $h_2^0(\bar{S}_2) = 0$, the L.H.S by definition is $\gamma_2(\bar{S}_2) = \mathbb{E} \left(Y_3^{A_0, A_1, 1} - Y_3^{A_0, A_1, 0} \mid \bar{S}_2 \right)$, and the R.H.S is also $\mathbb{E} \left(Y_3^{A_0, A_1, 1} - Y_3^{A_0, A_1, 0} \mid \bar{S}_2 \right)$. One can check the situation for $h_2^1(\bar{S}_2) = 0$, $h_2^0(\bar{S}_2) = 1$. Thus,

$$\begin{aligned}
& \mathbb{E} \left\{ R \Delta \gamma_2(\bar{S}_2) \mid \bar{S}_1 \right\} \\
&= \mathbb{E} \left\{ \Delta \gamma_2(\bar{S}_2) \mid \bar{S}_1, A_1 = h_1^0(\bar{S}_1) \right\} \\
&= \mathbb{E} \left\{ \mathbb{E} \left(Y_3^{A_0, A_1, h_2^1} - Y_3^{A_0, A_1, h_2^0} \mid \bar{S}_2 \right) \mid \bar{S}_1, A_1 = h_1^0(\bar{S}_1) \right\} \\
&= \mathbb{E} \left\{ Y_3^{A_0, h_1^0, h_2^1} - Y_3^{A_0, h_1^0, h_2^0} \mid \bar{S}_1, A_1 = h_1^0(\bar{S}_1) \right\} \\
&= \mathbb{E} \left\{ Y_3^{A_0, h_1^0, h_2^1} - Y_3^{A_0, h_1^0, h_2^0} \mid \bar{S}_1 \right\}.
\end{aligned}$$

With these results, the moment condition follows as in the classic SNMM literature. $\square$

REFERENCES

- Guo, F. R., Richardson, T. S., and Robins, J. M. (2021). Discussion of ‘estimating time-varying causal excursion effect in mobile health with binary outcomes’ by t. qian et al. *arXiv preprint arXiv:2103.02323*.
- Kennedy, E. H., Balakrishnan, S., and Wasserman, L. (2023). Semiparametric counterfactual density estimation. *Biometrika*, 110(4):875–896.
- Miguel, A., HERNAN, R., and James, M. (2023). *Causal inference: what if*. CRC PRESS.
- Robins, J. M. (2004). Optimal structural nested models for optimal sequential decisions. In *Proceedings of the Second Seattle Symposium in Biostatistics: analysis of correlated data*, pages 189–326. Springer.
- Robins, J. M., Rotnitzky, A., and Scharfstein, D. O. (2000). Sensitivity analysis for selection bias and unmeasured confounding in missing data and causal inference models. In *Statistical models in epidemiology, the environment, and clinical trials*, pages 1–94. Springer.